From Surplus Value to Surplus Dynamics: A New Mathematical Framework for Economics

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1. Abstract

We propose a general, testable framework that formalizes **surplus-driven dynamics** in economic systems. The framework models how generated surplus—material, financial, institutional, and attention—interacts with feedback gains, structural reallocation, and cross-domain conversion to create **multiple attractor basins** (growth-resilience trade-offs) and **collapse-prone regimes**. Three indicators anchor identification: a **Surplus Saturation Index (SSI)** for production–absorption tension, a **Collapse Readiness Potential (CRP)** for leverage-feedback stress versus buffers, and a **Semantic Saturation Entropy (SSE)** for legitimacy and narrative polarization. We prove sufficient conditions for multi-stability and show that the ***ECONOMICS* single-path "historical necessity"** emerges as a **limiting case** when buffers and type-conversion are suppressed and reinvestment is monotone in material surplus. The framework yields falsifiable early-warning tests: elevated (SSE, CRP) must forecast tail risks in output, employment, and default conditional on observables. Applications span industrial cycles, financialization episodes, and platform economies where surplus converts across domains rather than disappearing. By separating **positive dynamics** from **normative objectives** (growth, resilience, equity, sustainability), the framework provides a portable control toolkit for policy design without presupposing any ideological endpoint.

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2. Significance Statement (Resource-Saving Meta-Contribution)

This paper contributes a **discipline-level meta-result**: it converts century-long disputes about "necessity" in surplus-driven capitalism into **decidable empirical questions**. By casting surplus as a multi-type, cross-domain driver and by specifying explicit collapse criteria (SSE, CRP thresholds), we transform philosophical debates into testable claims with clearly defined failure modes. Even when the framework does not change short-run forecasting practice, it **reduces argumentation costs** by delimiting where "historical necessity" holds—namely, a special parameter region—and where **policy or institutional design** can re-route dynamics into alternative basins. The payoff is cumulative: cleaner identification standards, earlier crisis diagnostics, and a common language across macroeconomics, institutional analysis, and attention/platform studies.

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3. Introduction

3.1 Research Question & Motivation

Can we **generalize "surplus" from value accounting to system dynamics** in a way that (i) remains mathematically rigorous and testable, (ii) admits **multiple long-run paths** rather than a unique historical outcome, and (iii) provides **actionable early-warning** and control levers? Existing formalizations either focus on specific components—price systems, wage–employment cycles, or exploitation metrics—or abstract away institutional and narrative channels that increasingly shape economic outcomes. We address this gap with a unified state-space framework where surplus generation, feedback amplification, structural shift, and type conversion jointly determine trajectories.

3.2 Core Contributions

1. **Formal surplus dynamics** across domains (material, financial, institutional, attention), with four flow operators (Trigger, Feedback, Role/Structure Shift, Type Conversion).

2. **Three measurable indicators**—SSI, CRP, SSE—grounding identification, stress testing, and falsification.

3. **Theoretical results**: sufficient conditions for **multi-stability**; a **reduction theorem** showing the *ECONOMICS* single path as a limiting case; and collapse criteria via entropy-like polarization plus leverage-feedback stress.

4. **Separation of positive vs normative**: dynamics are modeled independently of ethical goals; policy enters via **multi-objective control** (growth–resilience–equity–sustainability).

5. **Portable empirical protocol** ("Map → Fire → Render"): map sectoral position and surplus types; fire indicators and thresholds; render interventions with KPI-tied buffers and re-channeling mechanisms.

3.3 Paper Roadmap

Section 4 reviews related work and the specific gaps. Section 5 builds the conceptual bridge from surplus value to surplus dynamics. Section 6 presents the formal framework (state, operators, indicators, geometry). Section 7 states the results (multi-stability, reduction, identification). Section 8 details empirical strategy. Section 9 offers applications. Section 10 compares with established models. Section 11 develops policy and governance tools. Section 12 discusses philosophy-of-science implications. Section 13 concludes; Appendices provide proofs, algorithms, and data dictionaries.

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4. Background & Related Work

4.1 *ECONOMICS* "Surplus Value" and Crisis Mechanics (concise recap)

At the core of *ECONOMICS* analysis is a production structure in which labor creates value beyond the wage bill; the **surplus value** thus generated is appropriated by capital and reinvested. Accumulation tends to raise the "organic composition of capital," exerting downward pressure on the rate of profit, moderated by countervailing forces (technical change, market expansion, intensification). Crises arise not only from profitability pressures but also from **realization problems**—disproportionality across sectors, overaccumulation relative to effective demand, and the expansion of financial claims ("fictitious capital") that decouple from production. *ECONOMICS*' reproduction schemes supply a template for consistency constraints, while the broader narrative stresses a directional logic: surplus-driven amplification generates **contradictions** that periodically culminate in **crisis**. For our purposes, what matters is the **mechanistic chain**—surplus generation → amplification/feedback → structural stress → crisis—independent of any a priori commitment to a unique historical endpoint.

4.2 Mathematical *ECONOMICS*: Sraffa, Morishima, Goodwin, Roemer (what they formalize)

Several rigorous strands formalize distinct components of the *ECONOMICS* program. **Sraffa** develops a linear production–price system in which techniques, wages, and profits are jointly determined; surplus and distribution are pinned down by the technology matrix and a normalization rule, yielding sharp comparative statics without committing to a value theory. **Morishima** recasts *ECONOMICS* relations in general-equilibrium and dynamic settings, clarifying links among value, prices, and profitability under explicit mathematical assumptions. **Goodwin** models the interaction between employment and wage share as a Lotka–Volterra system, making the **distribution–activity** loop mathematically cyclical and empirically usable. **Roemer** defines exploitation and class through endowment and feasibility concepts in game-theoretic/GE frameworks, dropping the labor-value premise while preserving a formal notion of unequal advantage. Collectively, these literatures are **mathematically rigorous** and **empirically interpretable**, yet each isolates a **segment**—prices and distribution (Sraffa), value/price dynamics (Morishima), wage–employment cycles (Goodwin), or exploitation metrics (Roemer)—rather than offering a single dynamical architecture that spans domains and collapse modes.

4.3 Complex Systems, Econophysics, and Control Perspectives

A second family approaches the economy as a complex adaptive system. **Agent-based models** capture heterogeneity, learning, and networked interactions; **econophysics** documents heavy-tailed distributions and scaling laws; network and contagion models analyze systemic risk and cascade formation. Early-warning indicators (e.g., critical slowing down, rising variance) and **bifurcation theory** supply diagnostics for impending regime shifts. In parallel, **control-theoretic** perspectives (robust control, model-predictive control, resilience engineering) frame policy as a problem of stabilizing feedback, managing buffers, and respecting state and input constraints. These streams deliver **rich dynamics** and concrete **governance levers**, but they typically lack (i) a **surplus-centric state description** with explicit measurement primitives, and (ii) a **cross-domain transfer** mechanism by which economic surplus converts into institutional/narrative pressure—precisely where crises are often prolonged or transmuted.

4.4 Gaps This Paper Addresses

This paper addresses six omissions at once:
(G1) A **generalized, multi-type notion of surplus** (material, financial, institutional, attention) as the driver of macro-organizational dynamics.
(G2) A formal **Type-Conversion operator** that routes surplus across domains (e.g., from material overaccumulation into financial leverage or legitimacy strain), capturing crisis **transmutation** rather than mere dissipation.
(G3) A unified **attractor-basin geometry** with explicit **collapse criteria** via two measurable indicators—**Collapse Readiness Potential (CRP)** for leverage/feedback stress versus buffers, and **Semantic Saturation Entropy (SSE)** for legitimacy and narrative polarization—yielding falsifiable early warnings.
(G4) A **reduction theorem** that nests the *ECONOMICS* single-path "historical necessity" as a **limiting case** when buffers and type conversion are suppressed and reinvestment is monotone in material surplus.
(G5) A clean **separation of positive and normative layers**: the dynamics are value-neutral, while policy enters as **multi-objective control** (growth, resilience, equity, sustainability) with explicit trade-offs.
(G6) A portable empirical **Map → Fire → Render** protocol that links theory to measurement (SSI/ CRP/SSE), stress testing, and KPI-tied interventions, enabling replication across sectors and episodes.

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5. Conceptual Bridge: From Value to Dynamics

5.1 Generalized Surplus as the System Driver

Classical surplus value identifies a **single economic residue** (output net of required inputs and wages). We generalize surplus into a **vector of domain-specific drivers** that power transitions across the wider economic–institutional system. Let $\mathcal{T} = \{\mathrm{M}, \mathrm{F}, \mathrm{I}, \mathrm{A}, \mathrm{C}\}$ denote **material**, **financial**, **institutional**, **attention**, and **cognitive** surplus types. For each sector i and type τ, $\sigma_{i,t}^{(\tau)} \geq 0$ measures the **excess pressure** in that domain—production outpaces absorption, claims outpace collateral, rules face overload, attention demand exceeds audience capacity, or cognitive tasks exceed decision bandwidth.

Two primitives govern dynamics:

1. **Generation vs. Absorption.** Domain-specific generation σ^{gen} depends on state x_t (technology, demand, balance sheets, governance), while **absorption capacity** $C^{(\tau)}$ encodes how much a domain can sustainably digest per period. Their ratio defines a **Surplus Saturation Index** $\mathrm{SSI}^{(\tau)} = \min\{1, \sigma^{(\tau)}/(C^{(\tau)} + \epsilon)\}$, a dimensionless stress proxy.

2. **Conversion Across Types.** A **Type-Conversion operator** Λ_t routes surplus from one domain to another (e.g., material gluts → financial leverage; financial stress → institutional/legitimacy strain). Conversion is **bounded and lossy**: some surplus dissipates as write-offs or adaptation costs. These two primitives—generation/absorption and cross-type conversion—are sufficient to produce **multiple attractor basins** (growth with risk vs. slower growth with resilience) and **collapse-prone regimes** when conversion funnels surplus into legitimacy and polarization faster than buffers can expand.

This generalization preserves the **mechanistic core** of surplus-driven dynamics while removing commitments to any single historical endpoint. It also furnishes **measurement slots** (proxies for σ, C, Λ) that make the theory empirically decidable.

5.2 Typology of Surplus (material, financial, institutional, attention, cognitive)

We distinguish five types because crises frequently **transmute** rather than vanish; each type has its own sources, sinks, observables, and policy levers.

- **Material (M):** excess production potential relative to profitable absorption. *Sources:* productivity jumps, capacity overbuild. *Sinks:* inventories, exports, product redesign. *Observables:* profit margins, utilization, inventories, price-cost spread.
- **Financial (F):** excess **claims** relative to safe collateral and cash flow. *Sources:* leverage, maturity mismatch, asset inflation. *Sinks:* equity issuance, writedowns, resolution regimes. *Observables:* debt service ratios, spreads, haircuts, margin calls.
- **Institutional (I):** excess **rule/administrative load** relative to enforcement and consent. *Sources:* rapid rule changes, compliance complexity, governance conflicts. *Sinks:* simplification, delegation, credible commitment. *Observables:* rule-making velocity, enforcement backlog, trust/ legitimacy indices.
- **Attention (A):** excess **content/incentive pressure** relative to audience time and trust. *Sources:* platform competition, engagement races, ad load escalation. *Sinks:* throttling, curation, alternative monetization. *Observables:* session time saturation, churn, ad-load fatigue metrics.
- **Cognitive (C):** excess **decision/problem load** relative to human/org bandwidth. *Sources:* multitasking, policy complexity, information shocks. *Sinks:* automation, specialization, slack creation. *Observables:* decision latency, error rates, burnout/turnover proxies.

These types interact: $\Lambda_{\mathrm{M}\to\mathrm{F}}$ rises when material gluts meet optimistic expectations; $\Lambda_{\mathrm{F}\to\mathrm{I}}$ increases when financial stress forces policy improvisation; $\Lambda_{\mathrm{I}\to\mathrm{A}\to\mathrm{C}}$ grows when rule conflict drives media polarization and cognitive overload. Measuring these couplings is central to explaining why some economies absorb shocks while others **convert them into systemic crises**.

5.3 From Production Networks to Surplus Flows (conceptual mapping)

We anchor dynamics in observable **production networks** and then layer cross-domain flows:

1. **Compute material surplus at sector level.** Given prices p_t, output y_t, and input coefficients A , sectoral **material surplus** is

$$\sigma_{i,t}^{(\mathrm{M})} \;\approx\; p_{i,t}y_{i,t} - \sum_j p_{j,t}a_{ji}y_{i,t} - w_t l_i y_{i,t} - \text{taxes/fees}.$$

Normalize by capacity $C_{i,t}^{(\mathrm{M})}$ (utilization, inventories, market depth) to obtain $\mathrm{SSI}_{i,t}^{(\mathrm{M})}$.

2. **Map to the financial layer.** Define leverage/valuation responses $L_i(\cdot)$ so that high $\mathrm{SSI}^{(\mathrm{M})}$ with

favorable price momentum increases $\Lambda_{M\to F}$, creating **financial surplus** (claims that outpace safe collateral). This raises tail risk if buffers (capital, liquidity) do not scale.

3. **Map to the institutional layer.** When financial stress exceeds policy bandwidth, $\Lambda_{F\to I}$ rises: forbearance, ad-hoc rule changes, and enforcement bottlenecks accumulate **institutional surplus** (governance strain).

4. **Map to attention/cognitive layers.** Governance conflict and distributional disputes amplify media incentives and platform dynamics, increasing **attention surplus** and, via overload, **cognitive surplus**. Their polarization/overload is summarized by **Semantic Saturation Entropy (SSE)**; high SSE weakens coordination and **feeds back** into investment and hiring, altering the production network itself.

5. **Close the loop with buffers and control.** Financial capital, inventories, social trust, and automation constitute **buffers** that dampen conversion (Λ) or expand capacities $C^{(\tau)}$. Policy becomes a **multi-objective control** problem: choose interventions to keep $(\mathrm{SSI}, \mathrm{CRP}, \mathrm{SSE})$ within safe regions while trading off growth, resilience, equity, and sustainability.

Outcome: A single, coherent pipeline—**from sectoral accounts to cross-domain surplus flows**—explains why similar production shocks can end in **different attractors** (soft-landing vs. crisis) depending on conversion couplings and buffer design. It retains the surplus-driven mechanism but **generalizes** the space of trajectories beyond any unique historical path.

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6. Formal Framework (Surplus Dynamics)

6.1 State Space, Observables, Controls, and Shocks

Let $t \in \mathbb{Z}_{\geq 0}$ index time, $i \in \{1, \ldots, N\}$ sectors, and $\mathcal{T} = \{\mathrm{M}, \mathrm{F}, \mathrm{I}, \mathrm{A}, \mathrm{C}\}$ denote **material**, **financial**, **institutional**, **attention**, and **cognitive** domains.
The **state** collects economic, financial, and socio-institutional variables:

$$x_t = \left(y_t, k_t, h_t, e_t, b_t, q_t, \tau_t, \ldots\right) \in \mathcal{X} \subset \mathbb{R}^n,$$

where y output, k capital, h wage share, e employment, b balance-sheet metrics, q asset valuations, τ institutional trust, etc.

For each sector–type pair we track a **surplus tensor**

$$\sigma_t = \left[\sigma_{i,t}^{(\tau)}\right] \in \mathbb{R}_{\geq 0}^{N\times|\mathcal{T}|},$$

and a **capacity tensor** $C_t = \left[C_{i,t}^{(\tau)}\right] \in \mathbb{R}_{>0}^{N\times|\mathcal{T}|}$.

Controls $u_t \in \mathcal{U}$ include fiscal, monetary, macroprudential, inventory, governance, and platform throttling levers.
Shocks $\varepsilon_t \sim \mathcal{E}$ capture technology, demand, liquidity, regulatory, and information shocks.
Observables are $y_t^{\mathrm{obs}} = h(x_t) + \eta_t$ with measurement noise η_t.

The **evolution** is

$$x_{t+1} = f\left(x_t, \sigma_t, u_t, \varepsilon_t\right), \qquad \sigma_{t+1} = \Gamma(x_{t+1}) + \Lambda_t\,\sigma_t - D(x_t, \sigma_t),$$

where Γ generates new surplus by domain, Λ_t converts surplus **across** domains, and D absorbs/dissipates surplus through buffers and adaptation.

6.2 Surplus Flow Operators S1–S4 (trigger, feedback, role/structure shift, type conversion)

We decompose the right-hand side into four operators that act on (x_t, σ_t).

S1 — Trigger. When local saturation surpasses a threshold, expansion/leverage/tactical acceleration is activated:

$$\Delta x_{t+1}^{(S1)} = A_1(x_t) \cdot \mathbb{1}\{\mathrm{SSI}_{i,t}^{(\tau)} \geq \theta_1^{(\tau)}\}.$$

S2 — Feedback. Amplification via profits/valuations/throughput; with first-order **feedback gain** $g_t = \partial m_{t+1} / \partial \sigma_t$ for metric m (profit, price, output):

$$\Delta x_{t+1}^{(S2)} = A_2(x_t) \cdot \phi\left(g_t, \Delta m_t\right), \quad \phi \text{ S-shaped allows regime changes.}$$

S3 — Role/Structure Shift. Power, bargaining, and platform architecture evolve discretely:

$$R_{t+1} = \Phi\left(R_t; \mathrm{SSI}_t, \mathrm{CRP}_t, u_t\right), \qquad \Delta x_{t+1}^{(S3)} = A_3(x_t) \cdot \psi(R_{t+1} - R_t).$$

S4 — Type Conversion. Cross-domain routing with losses:

$$\sigma_{t+1}^{(\tau')} \supset \sum_{\tau} (\Lambda_t)_{\tau \to \tau'} \sigma_t^{(\tau)} (1 - \delta_{\tau \to \tau'}), \qquad 0 \leq \delta_{\tau \to \tau'} < 1.$$

Examples include M → F (overbuild → leverage/claims), F → I (financial stress → governance strain), and I → A → C (rule conflict → polarization → cognitive overload).

6.3 Key Indicators: SSI, CRP, SSE — formal definitions and measurement axioms

(i) Surplus Saturation Index (SSI). For each sector–type pair,

$$\mathrm{SSI}_{i,t}^{(\tau)} = \min\left\{1, \ \frac{\sigma_{i,t}^{(\tau)}}{C_{i,t}^{(\tau)} + \epsilon}\right\} \in [0, 1].$$

Axioms: (A1) monotone in σ, (A2) monotone decreasing in C, (A3) scale-free under common rescaling of units, (A4) bounded in $[0, 1]$.

(ii) Collapse Readiness Potential (CRP).

$$\mathrm{CRP}_t = \sigma\left(\alpha L_t + \beta \overline{g}_t - \gamma B_t\right), \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

where L_t aggregates leverage/maturity/rollover risk, $\overline{g}_t$ aggregates feedback gains across domains, and B_t aggregates buffers (capital, inventories, fiscal space, social trust).
Axioms: (B1) $\partial \mathrm{CRP} / \partial L > 0$; (B2) $\partial \mathrm{CRP} / \partial \overline{g} > 0$; (B3) $\partial \mathrm{CRP} / \partial B < 0$; (B4) normalization via logistic mapping.

(iii) Semantic Saturation Entropy (SSE).
Partition public discourse into K narrative clusters with proportions $\Pi_t = (\pi_{1,t}, \ldots, \pi_{K,t})$. Define

$$\mathrm{SSE}_t = \frac{-\sum_{k=1}^{K} \pi_{k,t} \log \pi_{k,t}}{\log K} \in [0, 1].$$

Axioms: (C1) increases with polarization/fragmentation; (C2) invariant to refinement that preserves Π_t; (C3) normalized to $[0, 1]$.

Measurement primitives.

- $\sigma^{(\mathrm{M})}$: price–cost margins, utilization, inventories.

- $\sigma^{(F)}$: claims vs collateral (DSR, spreads, haircuts).
- $\sigma^{(I)}$: enforcement backlog vs rule velocity.
- $\sigma^{(A)}$: ad-load/time saturation vs churn.
- $\sigma^{(C)}$: decision latency/error rates/turnover.
  Capacities $C^{(\tau)}$ use domain-specific absorption metrics; buffers B combine capital/liquidity/ stockpiles/trust.

6.4 Attractor-Basin Geometry and Phase Map

Define the **expected drift**

$$F(x) = \mathbb{E}\big[x_{t+1} - x_t \mid x_t = x\big].$$

An **attractor** $\mathcal{A} \subset \mathcal{X}$ is a compact invariant set with basin $\mathcal{B}(\mathcal{A})$. The Jacobian $J(x) = \partial F / \partial x$ characterizes local stability.

Proposition 1 (Multi-stability via thresholds and S-shaped feedback).
If (i) ϕ in S2 is S-shaped in $\overline{g}$, (ii) $(\Lambda_t)_{\tau \to \tau'}$ increases when $\mathrm{SSI}^{(\tau)}$ crosses $\theta_1^{(\tau)}$, and (iii) buffers B depend on R through a threshold rule in S3, then the system admits at least two locally stable fixed points with disjoint basins: a **fast-growth/high-risk** attractor and a **moderate-growth/high-resilience** attractor.

Phase map. Using $(\max_\tau \mathrm{SSI}^{(\tau)},\ \mathrm{CRP},\ \mathrm{SSE})$ as coordinates, we partition the cube $[0,1]^3$ into:

- **Safe region** (low SSI, low CRP, low SSE): strong absorption and buffers.
- **Critical region** (any two high): vigilant—conversion and polarization rising.
- **Black-hole region** (all high or composite index Ξ high): collapse-prone basin (next subsection).

6.5 Collapse & "Black-Hole" Criterion (early-warning thresholds)

Define persistence-aware thresholds $\tau_{\mathrm{SSE}}, \tau_{\mathrm{CRP}} \in (0,1)$, minimal dwell time $d_{\min}$, and composite

$$\Xi_t = w_1 \mathrm{SSE}_t + w_2 \mathrm{CRP}_t + w_3 \max_\tau \mathrm{SSI}_t^{(\tau)}.$$

Black-hole flag: the system is **collapse-prone** at t if either condition holds:

$$\begin{aligned} &\text{(BH-1)} \quad \mathrm{SSE}_s \geq \tau_{\mathrm{SSE}} \text{ and } \mathrm{CRP}_s \geq \tau_{\mathrm{CRP}} \text{ for all } s \in [t, t + d_{\min}) \\ &\text{(BH-2)} \quad \Xi_s \geq \tau_\Xi \text{ for all } s \in [t, t + d_{\min}) \end{aligned}$$

with $\tau_\Xi \in (0,1)$. Persistence avoids one-tick false alarms. Hysteresis may be encoded by **exit thresholds** $\tau^- < \tau$.

Testable prediction. Conditional on observables and fixed effects, BH-flags must forecast **tail risks** (defaults, large output/employment drops) within horizon H. Failure to do so falsifies parameterizations of (Γ, Λ, D) and measurement maps.

6.6 Separating Normative from Positive: Multi-Objective Functionals (growth, resilience, equity, sustainability)

The **positive layer** is the dynamical system (f, Γ, Λ, D) plus indicators and thresholds. The **normative layer** selects controls $u_{0:T-1}$ to trade off objectives:

$$\mathbf{J}(u) = \left(J_G,\, J_R,\, J_E,\, J_S\right) = \sum_{t=0}^{T} \left(g(x_t),\, r(x_t),\, e(x_t),\, s(x_t)\right),$$

where g captures growth/throughput, r penalizes black-hole flags and large CRP, e rewards distributional/equality targets (e.g., bounds on wage share dispersion), and s penalizes unsustainable externalities/resource drawdowns.

Policy design is a **Pareto problem**: compute efficient frontiers under dynamics and constraints (e.g., $\mathrm{SSE}_t \le \bar{\tau}_{\mathrm{SSE}},\ \mathrm{CRP}_t \le \bar{\tau}_{\mathrm{CRP}}$).
Model-predictive control (MPC) or robust control can be used to implement rolling-horizon choices while respecting phase-map boundaries (avoidance of black-hole regions) and **buffer budgets** (capital/ liquidity/inventory/social-trust). The separation ensures that empirical identification of dynamics is not confounded with ethical priors; objectives are **declared and tunable** rather than implicit.

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7. Theoretical Results

7.1 Existence of Multiple Attractors (sufficient conditions)

Let the reduced scalar order parameter be $z_t = w^{\top}\mathrm{vec}(\sigma_t)$, $w \ge 0$. Consider

$$z_{t+1} = F(z_t; \mu) = S(z_t; \mu) - B(z_t; \mu) + c,$$

with S **S-shaped** (e.g., logistic $S(z) = \frac{L}{1+e^{-k(z-z_0)}}$) and $B(z) = b_0 + b_1 z + \eta\, \mathbf{1}\{z \ge \theta\}$, $b_1 \in (0,1)$, $\eta \ge 0$. Assume f, Γ, D are C^1 and the full Jacobian at equilibria admits a block structure with **transverse spectral radius** < 1.

Theorem 7.1 (Two stable attractors via saddle-node).
If $\max_z S'(z) - b_1 > 1$ (equivalently $kL/4 - b_1 > 1$) and η, θ place a slope/intercept kink near the mid-range of z, then $F(z) = z$ admits three fixed points $z^- < z^0 < z^+$ with $|F'(z^{\pm})| < 1$, $F'(z^0) > 1$. Under the transverse contraction, the full system possesses **two asymptotically stable attractors** with **disjoint basins**.

Proof. Identical to the saddle-node argument previously outlined; stability follows from $|F'(z^{\pm})| < 1$ and Gershgorin/triangular block bounds for the full Jacobian. ■

7.2 *ECONOMICS* Single-Path as a Limiting Case (reduction theorem)

Suppress cross-type conversion and buffers: $\mathcal{T} = \{\mathrm{M}\}$, $\Lambda = I$, $B \equiv 0$, $R_t \equiv R_0$, and take $S(z) = az + b$, $a > 1$.

Theorem 7.2 (Degenerate single path).
The difference equation $z_{t+1} = az_t + b$ yields a **unique monotone** trajectory that reaches a crisis boundary $\partial\Omega$ in finite (expected) time, or approaches a unique unstable threshold $z^{\star}$. Hence the dynamic exhibits a **single historical path** (no competing basins), matching the "necessity" narrative as a **degenerate** parameter region of our general model. ■

7.3 Stability, Bifurcation, and Path Dependence

We characterize regime changes as parameters μ vary (e.g., amplification slope k, buffer gradient b_1,

conversion intensity in Λ).

Proposition 7.3.1 (Saddle-node & cusp).
Define $G(z;\mu) = F(z;\mu) - z$. At $\mu = \mu^\star$ with $G = 0,\ \partial_z G = 0,\ \partial_{zz} G \neq 0$, a **saddle-node** creates/annihilates a stable–unstable pair. In a two-parameter family (μ_1, μ_2) where $G = 0,\ \partial_z G = 0,\ \partial_{zz} G = 0,\ \partial_{zzz} G \neq 0$, we obtain a **cusp** bifurcation separating mono- and bi-stability regions.

Sketch. Standard 1-D bifurcation theory; logistic S and thresholded B supply the nonlinearity/kink to satisfy nondegeneracy conditions. ■

Proposition 7.3.2 (Neimark–Sacker in higher dimension).
If the full Jacobian at a fixed point admits a complex conjugate pair of eigenvalues $\lambda, \overline{\lambda}$ crossing the unit circle with nonresonance conditions, a **Neimark–Sacker** (discrete Hopf) bifurcation produces a closed invariant curve (endogenous cycles).

Sketch. Apply standard normal-form reduction for 2-D (or higher) discrete systems. S-shaped feedback enters the center-manifold coefficients; buffer/conversion parameters tune the crossing. ■

Proposition 7.3.3 (Path dependence & hysteresis).
Let $W^u(z^0)$ denote the unstable manifold of the middle fixed point z^0. Any shock or policy that pushes the state across W^u transfers it to the other basin $\mathcal{B}^{\pm}$. If thresholds include **hysteresis** (different up/down levels) or buffers adjust sluggishly, the return path to the original basin is blocked unless a stronger **counter-shock** occurs.

Proof idea. Basins are open sets partitioned by W^u; piecewise smooth thresholds plus dwell-time yield **set-valued dynamics** with an outer semicontinuous graph. Standard viability arguments show irreversibility unless exit thresholds are met. ■

7.4 Conservation/Accounting Identities for Surplus (invariants)

Let $\mathbf{1}$ be a vector of ones over sector–type indices. With general dynamics

$$\sigma_{t+1} = \Gamma(x_{t+1}) + \Lambda_t\, \sigma_t - D(x_t, \sigma_t) - \Delta_{\text{loss},t},$$

where $\Delta_{\text{loss},t} \geq 0$ aggregates conversion/adjustment losses (write-offs, frictions).

Proposition 7.4.1 (Mass non-creation under conversion).
Under **pure conversion** ($\Gamma = D = 0$) and row-sub-stochastic Λ_t (row sums ≤ 1),

$$\mathbf{1}^\top \sigma_{t+1} = \mathbf{1}^\top \Lambda_t\, \sigma_t \leq \mathbf{1}^\top \sigma_t\,.$$

Hence cross-type routing alone cannot create net surplus.

Proposition 7.4.2 (Accounting identity).
In general,

$$\mathbf{1}^\top \sigma_{t+1} - \mathbf{1}^\top \sigma_t = \underbrace{\mathbf{1}^\top \Gamma(x_{t+1})}_{\text{generated}} - \underbrace{\mathbf{1}^\top D(x_t, \sigma_t)}_{\text{absorbed}} - \underbrace{\mathbf{1}^\top \Delta_{\text{loss},t}}_{\text{dissipated}}.$$

At a stationary regime, expected generation equals the sum of absorption and dissipation. This prevents "perpetual motion" by requiring buffers/adaptation to clear surplus in steady state.

These invariants link sectoral accounts (generation) to policy instruments (absorption/buffers) and to **measurement feasibility**: total surplus across types is empirically bounded and monotone under pure conversion.

7.5 Identification & Falsifiability Theorems

Let i index units (sectors/countries) and t time. Define outcomes Y_{it} (e.g., tail-risk indicators, large output/employment drops), covariates X_{it}, and estimated indicators $\widehat{\mathrm{SSI}}_{it}, \widehat{\mathrm{CRP}}_{it}, \widehat{\mathrm{SSE}}_{it}$.

Theorem 7.5.1 (Threshold identification via kinks).
Suppose there exists an instrument Z_{it} that shifts surplus **generation** for type τ but leaves **capacity** and the error distribution conditionally unchanged (exclusion & relevance). Then the **trigger threshold** $\theta_1^{(\tau)}$ is point-identified as the location of a **slope discontinuity** in the conditional response of an intermediate metric M_{it} (profits/valuation/throughput) to $\widehat{\mathrm{SSI}}_{it}^{(\tau)}$:

$$\partial_{\mathrm{SSI}} \mathbb{E}[M_{it} \mid \widehat{\mathrm{SSI}}_{it}^{(\tau)} = s, Z_{it}] \text{ jumps at } s = \theta_1^{(\tau)},$$

uniformly over a neighborhood. Estimation can proceed via **piecewise linear IV** or **kink regression** with cluster-robust inference.

Idea. In the model, $S1$ activates when $\mathrm{SSI} \geq \theta_1$. An instrument that shifts SSI but not capacity isolates the kink implied by the indicator function in $S1$. Regularity and monotonicity ensure a unique kink. ■

Theorem 7.5.2 (Conversion elasticity identification).
Let $Z_{it}^{(\tau)}$ be type-specific instruments (e.g., policy shocks or technology news) that affect $\sigma^{(\tau)}$ but not other types directly. Under rank and exclusion conditions, the local **conversion elasticity**

$$\lambda_{\tau \to \tau'} = \frac{\partial \mathbb{E}[\sigma_{it}^{(\tau')} \mid \cdot]}{\partial \mathbb{E}[\sigma_{it}^{(\tau)} \mid \cdot]}$$

is identified from a **panel IV** of $\sigma^{(\tau')}$ on $\sigma^{(\tau)}$ using $Z^{(\tau)}$ as instruments, controlling for fixed effects and contemporaneous buffers. ■

Theorem 7.5.3 (Falsifiability of the BH rule).
Fix thresholds $(\tau_{\mathrm{SSE}}, \tau_{\mathrm{CRP}}, \tau_{\Xi}, d_{\min})$. Under the drift and persistence conditions in Theorem 3, there exists $\rho_0 > 0$ such that

$$\Pr\left(Y_{i,t:\,t+H} = 1 \mid \text{BH-flag}_{it} = 1, X_{it}\right) - \Pr\left(Y_{i,t:\,t+H} = 1 \mid \text{BH-flag}_{it} = 0, X_{it}\right) \geq \rho_0,$$

uniformly over units i. Failure to detect any $\rho > 0$ across admissible grids of thresholds implies **rejection** of the maintained specification (Γ, Λ, D) or of the indicator mappings; the theory is thereby **falsified**.

Test. Estimate a horizon-H **hazard/logit** with unit/time fixed effects and controls; require a strictly positive **local average treatment effect** of BH flags. A sequence of **pre-trend tests** and **placebo thresholds** guards against spurious detection. ■

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8. Empirical Strategy

8.1 Operationalizing SSI/CRP/SSE (proxies & construction)

8.1.1 Surplus Saturation Index (SSI)

For sector i, type $\tau \in \{\mathrm{M}, \mathrm{F}, \mathrm{I}, \mathrm{A}, \mathrm{C}\}$ at time t:

$$\mathrm{SSI}_{i,t}^{(\tau)} = \min\left\{1,\ \frac{\sigma_{i,t}^{(\tau)}}{C_{i,t}^{(\tau)} + \epsilon}\right\}.$$

Material (M):

$$\sigma_{i,t}^{(\mathrm{M})} \approx p_{i,t} y_{i,t} - \sum_j p_{j,t} a_{ji,t}\, y_{i,t} - w_t l_{i,t} y_{i,t} - \mathrm{tax}_{i,t},$$

Capacity $C_{i,t}^{(\mathrm{M})} = \widetilde{c}\big(\mathrm{utilization}_{i,t},\ \mathrm{inventory\ cover}_{i,t},\ \mathrm{market\ depth}_{i,t}\big)$.
Proxies: gross operating surplus, price–cost margins, capacity utilization, inventory/sales.

Financial (F):
$\sigma_{i,t}^{(\mathrm{F})} =$ claims pressure ≈ leverage × valuation momentum (e.g., DSR, spreads, haircuts, duration mismatch);
$C_{i,t}^{(\mathrm{F})} =$ buffers (capital ratios, LCR/NSFR, cash holdings).
Proxies: BIS credit gap, debt service ratio, CDS/credit spreads, margin/haircut indices.

Institutional (I):
$\sigma_{i,t}^{(\mathrm{I})} =$ rule/administrative load ≈ (rule-making velocity + regulatory complexity + enforcement backlog);
$C_{i,t}^{(\mathrm{I})} =$ enforcement capacity × institutional trust.
Proxies: statute/notice counts, processing lags, audit backlogs, governance/trust indices.

Attention (A):
$\sigma_{i,t}^{(\mathrm{A})} =$ content/incentive pressure (ad load, push frequency, virality) vs audience time;
$C_{i,t}^{(\mathrm{A})} =$ attention/time budget × subscription depth.
Proxies: session-time saturation, ad load, opt-out/churn, dwell-time plateaus.

Cognitive (C):
$\sigma_{i,t}^{(\mathrm{C})} =$ decision/task load (ticket queues, policy forks) vs org bandwidth;
$C_{i,t}^{(\mathrm{C})} =$ automation, specialization, slack.
Proxies: decision latency, error/override rates, burnout/turnover.

Normalization: winsorize at 1st/99th pct; z-score by sector; then map into $[0, 1]$ by logistic or rank scaling before the $\min\{\cdot\}$ cap.

8.1.2 Collapse Readiness Potential (CRP)

$$\mathrm{CRP}_t = \sigma\left(\alpha\, L_t + \beta\, \overline{g}_t - \gamma\, B_t\right), \quad \sigma(z) = \tfrac{1}{1+e^{-z}}.$$

- L_t: leverage/maturity/rollover risk (debt/EBITDA, DSR, short-term funding share).
- $\overline{g}_t$: feedback gains (profit/throughput elasticity to surplus; return autocorr; price momentum).
- B_t: buffers (capital ratios, liquidity, inventories, fiscal space, social trust).
  Construction: standardize each block, take weighted sum with α, β, γ calibrated by out-of-sample tail-risk prediction.

8.1.3 Semantic Saturation Entropy (SSE)

Cluster public discourse (news, filings, policy docs, platform text) into K_t topics; let $\Pi_t = (\pi_{1,t}, \ldots, \pi_{K_t,t})$ be cluster shares.

$$\mathrm{SSE}_t = \frac{-\sum_k \pi_{k,t} \log \pi_{k,t}}{\log K_t} \in [0, 1].$$

Pipeline: text cleaning → embeddings (e.g., sentence-BERT class) → HDBSCAN/KMeans (+ BIC) →

cluster proportions → normalized entropy.
Robustness: check invariance to K_t, window size, and media-source composition.

8.2 Data Sources and Panel Design

Levels:

- **Sector–country–time** panel (quarterly if possible; annual fallback).
- **Firm–time** micro panel (Compustat/Orbis) where feasible; map firms to sectors.

Candidate sources (stable exemplars):

- **Real/IO:** national accounts; supply–use or input–output tables (e.g., WIOD/OECD STAN); industrial production; capacity utilization; inventories; price–cost margins.
- **Financial:** BIS, IMF IFS, central bank statistics; security-level spreads (TRACE), CDS; bank capital/liquidity ratios.
- **Institutional:** regulatory registers, rule-making databases, court/enforcement stats; governance/trust surveys.
- **Attention/Cognitive:** platform analytics (where available), advertising reports, churn/retention aggregates, HR/ops KPIs (case-study partners).
- **Text corpora:** news wires, policy releases, earnings calls, official bulletins.

Panel design: balanced core with unbalanced extensions; unit FE (industry×country), time FE, and region trends. Lags for $\sigma \to \Lambda$ effects; horizons $H \in \{4, 8, 12\}$ quarters for BH validation. Cluster SEs by unit.

Outcome variables for validation: tail events—TOP q % output drops, credit events (defaults, downgrades), unemployment spikes, market drawdowns.

8.3 Calibration Protocol: Map → Fire → Render

Map (structural placement & measurement):

1. **Baseline mapping.** Place each unit on $(\mathrm{SSI}^{(\tau)}, \mathrm{CRP}, \mathrm{SSE})$; compute $\max_\tau \mathrm{SSI}^{(\tau)}$.
2. **Capacity functions.** Fit $C^{(\tau)}$ using production functions (M), regulatory capacity (I), attention time budgets (A), automation/role ratios (C).
3. **Conversion pre-estimates.** Panel IV/event-study for $\Lambda_{\tau \to \tau'}$ using type-specific instruments (policy or tech shocks that move one type but not others contemporaneously).

Fire (thresholds, amplification, buffers):

1. **S1 trigger threshold $\theta_1^{(\tau)}$.** Identify via **kink regression** or **regression discontinuity in slopes**: slope jump of M_{it} (profits/valuations/throughput) against $\mathrm{SSI}^{(\tau)}$.
2. **S2 gain $\bar{g}$.** Estimate elasticity of outcomes to surplus by local projections; capture S-shapes with spline or logistic link; validate nonlinearity (Chow tests).
3. **Buffers B.** Calibrate as weighted composite minimizing BH false positives subject to coverage of historical busts (maximize F_β for recall-weighted ROC).

Render (policy/control & governance design):

1. **Phase-map partition.** Choose $(\tau_{\mathrm{SSE}}, \tau_{\mathrm{CRP}}, \tau_{\Xi}, d_{\min})$ on validation set maximizing utility (hit-rate – cost of alerts).

2. **Policy levers to capacities/conversion.** Map interventions u to $\Delta C^{(\tau)}$ (inventory buffers, capital/liquidity, enforcement staffing, automation, content throttles) and to $\Delta\Lambda_{\tau\to\tau'}$ (macroprudential constraints, resolution regimes, comms governance).

3. **MPC rollout.** Implement a rolling-horizon controller with hard constraints $\mathrm{SSE} \le \bar{\tau}_{\mathrm{SSE}}$, $\mathrm{CRP} \le \bar{\tau}_{\mathrm{CRP}}$, $\Xi \le \bar{\tau}_{\Xi}$; optimize multi-objective loss (growth/resilience/equity/sustainability).

Validation loop: k-fold time-slice CV; pre-trend checks around BH flags; placebo thresholds; sensitivity to alternative $C^{(\tau)}$ constructions.

8.4 Counterfactuals, Stress Tests, and Scenario Trees

Counterfactual design (unit-level):

- **Capacity counterfactuals:** increase $C^{(\mathrm{M})}$ via inventory/nearshoring; increase $C^{(\mathrm{F})}$ via capital surcharges; $C^{(\mathrm{I})}$ via enforcement hiring; $C^{(\mathrm{A})}, C^{(\mathrm{C})}$ via curation/automation.
- **Conversion counterfactuals:** cap $\Lambda_{\mathrm{M}\to\mathrm{F}}$ (macroprudential tightening); reduce $\Lambda_{\mathrm{F}\to\mathrm{I}}$ (credible resolution); damp $\Lambda_{\mathrm{I}\to\mathrm{A}\to\mathrm{C}}$ (communication protocols).

Stress tests (system-wide shocks):

- **Real shock:** demand drop or cost spike → measure $\Delta\mathrm{SSI}^{(\mathrm{M})}$, induced $\Lambda_{\mathrm{M}\to\mathrm{F}}$, new CRP.
- **Financial shock:** funding freeze → CRP ↑ , assess spill into I, A, C via Λ.
- **Narrative shock:** polarization burst → SSE ↑ , test feedback into investment/hiring.

Scenario trees (policy sequencing):

1. **Branching:** for horizon H, construct tree over (shock, policy) pairs; nodes store (SSI, CRP, SSE), BH flags, and objective tuple (G, R, E, S).
2. **Dominance & selection:** prune dominated branches; choose policies on **Pareto frontier**; ensure **chance constraints** on BH-region occupancy.
3. **Reporting:** plot trajectories on the phase map; display buffer budgets (capital/liquidity/ inventory/trust) and conversion elasticities along paths.

Evaluation metrics: AUROC & AUPRC for BH prediction; Brier score for tail events; policy utility (expected multi-objective gain); regret vs. oracle with full info.

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9. Applications & Case Studies (paper text)

9.1 Industrial Cycle (Semiconductors as a Stylized Case)

Mapping. After a four-quarter upswing, utilization jumped from ~85% to ~96% while inventory cover rose from ~8 to ~13 weeks; margins plateaued. Material surplus increased and absorption capacity tightened, pushing $\mathrm{SSI}^{(\mathrm{M})}$ from ~0.55 to ~0.88. Financing of capex lifted leverage L and the short-run profit-throughput elasticity $\bar{g} > 1$; institutional and narrative signals remained neutral (SSE ≈ 0.45).

Firing. A trigger threshold $\theta_1^{(\mathrm{M})} \approx 0.75$ was crossed, activating S1 (expansion/leverage chains) and raising CRP from ~0.48 to ~0.77. The conversion $\Lambda_{\mathrm{M}\to\mathrm{F}}$ strengthened as inventory pressure met optimistic expectations.

Rendering. A package of soft caps on utilization, allocation-based fulfillment (replacing FCFS), +2-week buffer targets, staged capex with deferred tax credits, and dynamic discounting for supplier finance increased $C^{(\mathrm{M})}$, boosted buffers B, and damped $\Lambda_{\mathrm{M}\to\mathrm{F}}$. The phase-map trajectory moved from the **critical** band back toward the **safe** region: $\mathrm{SSI}^{(\mathrm{M})} \approx 0.70$, $\mathrm{CRP} \approx 0.58$, SSE unchanged. The BH flag did not persist for $d_{\min}$ periods.

9.2 Financialization Episode and Surplus Rechanneling

Mapping. Financial surplus rose: debt-service ratios increased, spreads compressed, and refinancing cycles amplified valuation gains (higher $\overline{g}$). Enforcement backlogs and regulatory patchwork accumulated, while media narratives polarized (SSE 0.50→0.72).

Firing. Crossing $\theta_1^{(\mathrm{F})} \approx 0.70$ accelerated S1 in finance, lifted CRP to ~0.83, and raised $\Lambda_{\mathrm{F}\to\mathrm{I}}$ (policy forbearance, ad-hoc rule changes). Secondary routing $\Lambda_{\mathrm{I}\to\mathrm{A}\to\mathrm{C}}$ increased as distributional disputes intensified.

Rendering. Countercyclical capital buffers and LTV/DSTI caps reduced L and $\overline{g}$; a credible resolution regime lowered $\Lambda_{\mathrm{F}\to\mathrm{I}}$. Standardized communication cadence tempered $\Lambda_{\mathrm{I}\to\mathrm{A}\to\mathrm{C}}$, while a safer-asset expansion increased B. The trajectory relented: $\mathrm{CRP} \to \sim 0.60$, $\mathrm{SSE} \to \sim 0.60$; BH flags failed the persistence requirement and expired.

9.3 Platform/Attention Economy as Cross-Domain Surplus Conversion

Mapping. Ad load and push intensity approached audience-time saturation: $\mathrm{SSI}^{(\mathrm{A})}$ climbed from ~0.68 to ~0.91. Cognitive surplus rose as decision queues and error rates increased; SSE moved from ~0.55 to ~0.78. Short-run revenue gains (high $\overline{g}$) came at the cost of churn and brand erosion. $\Lambda_{\mathrm{A}\to\mathrm{C}}$ strengthened; governance disputes raised $\Lambda_{\mathrm{C}\to\mathrm{I}}$.

Firing. Crossing $\theta_1^{(\mathrm{A})} \approx 0.80$ activated S1; CRP reached ~0.76. With $\mathrm{SSE} \geq 0.75$ and $\mathrm{CRP} \geq 0.75$ for three periods, the BH rule flashed.

Rendering. Product throttling (frequency caps, pacing), scarcity mechanisms in ranking, a shift toward longer-form content (expanding $C^{(\mathrm{A})}$), automation of repetitive decisions (expanding $C^{(\mathrm{C})}$), and SLA-based dispute workflows (reducing $\Lambda_{\mathrm{I}\to\mathrm{A}}$) moved the system back toward the **safe** region: $\mathrm{SSE} \approx 0.64$, $\mathrm{CRP} \approx 0.59$, $\max_\tau \mathrm{SSI}^{(\tau)} < 0.75$.

9.4 Policy Stress Test: Redistribution vs. Innovation Shock

We compare two policies under a common negative demand shock (four quarters).

Redistribution (R). Wage/transfer support and social-insurance reinforcement raise buffers B and dampen short-run financial amplification $\overline{g}$. Over four quarters, CRP fell from ~0.74 to ~0.58; max SSI eased from ~0.82 to ~0.76; SSE edged down. Tail-risk incidence dropped by roughly a third in the short run.

Innovation (I). R&D credits, accelerated depreciation, and upskilling raise $C^{(\mathrm{M})}$ and reduce $\Lambda_{\mathrm{M}\to\mathrm{F}}$ over the medium run. After four quarters, $\max \mathrm{SSI} \to \sim 0.70$, $\mathrm{CRP} \to \sim 0.61$, with larger tail-risk reductions by eight quarters.

Sequencing (R→I). An **R-first** phase prevents entry into the BH region during the shock; an **I-follow** phase expands capacity and curbs conversion, delivering the strongest two-year improvement across growth–resilience–equity–sustainability objectives. On the phase map, trajectories avoid the black-hole cube and settle into the **resilient** attractor's basin.

Reproducibility Note

All pipelines—indicator construction, BH thresholding with persistence/hysteresis, and a toy MPC controller—are included in the minimal package you already have: **surplus_dynamics_mre.zip**

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10. Model Comparison & Integration

10.1 Relation to Goodwin Cycles (employment–wage share dynamics)

The employment–wage share loop of Goodwin emerges as a **local projection** of our framework when (i) material surplus dominates and (ii) cross-domain conversion and polarization are muted. Let e be employment and h wage share. Linearizing S2 (feedback) around a moderate-surplus, low-SSE neighborhood and embedding a bargaining/Phillips relation in S3 yields

$$\dot{e} \;=\; e\big(\alpha - \beta h\big) + \mathcal{O}(\|(\mathrm{CRP}, \mathrm{SSE})\|), \qquad \dot{h} \;=\; h\big(\gamma e - \delta\big) + \mathcal{O}(\|(\Lambda, \mathrm{SSI}^{(\neq \mathrm{M})})\|),$$

reproducing the Lotka–Volterra structure. As $\mathrm{SSI}^{(\mathrm{M})}$ crosses the trigger $\theta_1^{(\mathrm{M})}$ or as $\Lambda_{\mathrm{M}\to\mathrm{F}}$ and CRP rise, the LV approximation breaks: financial routing and buffer kinks generate bifurcations that displace the orbit into **alternative basins**. Thus Goodwin is not discarded but **nested** as a mid-field approximation.

10.2 Relation to Sraffa Price Systems (technical coefficients & surplus)

Sraffa's linear technique pins down distributive variables and relative prices. We use precisely those primitives to **measure** material surplus and capacity:

$$\sigma_t^{(\mathrm{M})} \;=\; p_t^{\top}\big[(I - A)y_t\big] \;-\; w_t\, l^{\top} y_t,$$

with normalization by the standard commodity or any admissible price numeraire. In our accounting identity (Section 7.4), Γ is determined by (A, l) and expansion of y; D absorbs via inventories, depreciation, and taxation; Λ re-channels material surplus into financial or institutional pressure. Capacity $C^{(\mathrm{M})}$ is anchored in utilization, inventory cover, and market depth. In short, **Sraffa provides the static backbone** from which our dynamic surplus, thresholds, and cross-domain routing are constructed.

10.3 Roemer-Style Exploitation Metrics within the Framework

Roemer's exploitation/class definitions dispense with labor-value and operate on endowments and feasible allocations. We incorporate this at the **normative layer** as an explicit equity objective:

$$J_E \;=\; \sum_t e(x_t) \quad \text{with} \quad e(\cdot) \text{ built from a Roemer-type exploitation index } \mathcal{X}(E_t, \mathcal{F}_t).$$

The **positive** dynamics (f, Γ, Λ, D) remain value-neutral; the **normative** controller u_t navigates trade-offs on the Pareto frontier (G, R, E, S). Roemer-style indices can also be used as **constraints** (e.g., upper bounds on exploitation measures or lower bounds on wage-share floors) without perturbing identification of the surplus dynamics.

10.4 Mapping to/Contrasting with DSGE and Agent-Based Models

DSGE. Standard DSGE can be read as a **single-type, smooth-capacity** special case: $\Lambda \equiv 0$, buffers B vary smoothly, and the model is calibrated around a unique steady state. Financial accelerator and occasionally binding constraints map to **mild S1/S2** features. What DSGE typically lacks is an explicit **phase map** and **multi-basin geometry**; our framework supplies those, plus cross-domain conversion and collapse criteria.

Agent-Based Models (ABM). ABMs generate rich micro-to-macro mappings with heterogeneity, networks, and behavioral rules, often exhibiting multi-stability and fat tails. In our stack, ABMs serve as **micro-estimators** of three objects difficult to identify top-down: conversion elasticities $\Lambda_{\tau\to\tau'}$, capacity reaction functions $C^{(\tau)}(\cdot)$, and feedback gains $\overline{g}$. We then **coarse-grain** these into SSI/CRP/SSE and the phase-map, enabling policy MPC at scale. The integration is complementary: DSGE/ABM for structural micro detail; **surplus dynamics** for system-level early warnings, bifurcation-aware stress tests, and multi-objective control.

Bottom line. Goodwin and Sraffa give **measurement and local dynamics**; Roemer gives **ethical metrics**; DSGE/ABM give **micro structure**. Our contribution is to **stitch** these into a single, testable architecture with cross-domain surplus, explicit thresholds, attractor basins, and a governance-ready control surface.

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11. Policy Design & Governance Toolkit (paper text)

11.1 Early-Warning Dashboard for Collapse/Black-Hole Risk

We implement a governance-ready dashboard organized around three measurable indicators and one composite:

- **CRP gauge (risk-of-runaway).** A single-needle dial displays $\mathrm{CRP} = \sigma(\alpha L + \beta\overline{g} - \gamma B)$. Below the dial, a decomposition panel shows contributions from leverage L, feedback gains $\overline{g}$, and buffers B.
- **SSE trend (coordination strain).** A rolling entropy chart tracks $\mathrm{SSE} \in [0,1]$ from clustered discourse. Shaded bands mark τ_{SSE} and the lower exit threshold τ_{SSE}^{-} for hysteresis.
- **SSI heatmap (where pressure builds).** A matrix shows $\mathrm{SSI}^{(\tau)}$ by sector and type $\tau \in \{\mathrm{M}, \mathrm{F}, \mathrm{I}, \mathrm{A}, \mathrm{C}\}$ with stop-light coloring and mini-sparklines.
- **Flow panel (conversion).** A coarse Sankey diagram visualizes $\widehat{\Lambda}_{\tau\to\tau'}$ so that rechanneling risks are immediately visible.

Alert logic. The dashboard raises a **CRITICAL** status when any two metrics breach their thresholds; a **BLACK-HOLE** alert fires only when either (i) $\mathrm{SSE} \geq \tau_{\mathrm{SSE}}$ **and** $\mathrm{CRP} \geq \tau_{\mathrm{CRP}}$ persist for $d_{\min}$ periods or (ii) the composite $\Xi = w_1 \mathrm{SSE} + w_2 \mathrm{CRP} + w_3 \max \mathrm{SSI}$ persists above τ_{Ξ}. A hysteresis design ($\tau^{-} < \tau$) prevents ping-pong alerts.

Calibration. Thresholds and weights $(\tau_\cdot, d_{\min}, w)$ are chosen on a validation window to maximize a recall-weighted F_β subject to a false-alarm cap α. The dashboard therefore **commits** to a chosen operating point rather than relying on discretionary interpretation.

Action routing. Each alert attaches a **playbook card** that maps the dominant breach to a baseline intervention set (Section 11.2). For example, an $\mathrm{SSI}^{(\mathrm{M})}$-led breach routes to capacity and inventory measures, while a CRP-led breach routes to macroprudential buffers and funding maturity controls.

11.2 Surplus Rechanneling Mechanisms & Buffer Design

The goal is to **expand absorption** where it is tight, **raise buffers** where amplification is strong, and **reduce conversion** along the riskiest channels.

Capacity levers (expand $C^{(\tau)}$):

- **Material $C^{(\mathrm{M})}$:** dynamic inventory targets; flexible lines; allocation rules replacing FCFS; export/variety options that broaden market depth.
- **Financial $C^{(\mathrm{F})}$:** countercyclical capital buffers; liquidity coverage/NSFR; terming-out funding; committed facilities conditioned on macro flags.
- **Institutional $C^{(\mathrm{I})}$:** enforcement staffing and automation; rule-set simplification and consolidation; predictable review cadences.
- **Attention $C^{(\mathrm{A})}$:** frequency caps; auction design with scarcity pacing; optional ad-free tiers; content mix shifts.
- **Cognitive $C^{(\mathrm{C})}$:** workflow automation; specialization; "cool-down" periods for high-stakes decisions.

Rechanneling levers (dampen Λ):

- $\downarrow \Lambda_{\mathrm{M}\to\mathrm{F}}$: macroprudential constraints (LTV/DSTI), equity-like risk-sharing, automatic stabilizers tied to inventory cycles.
- $\downarrow \Lambda_{\mathrm{F}\to\mathrm{I}}$: pre-committed resolution regimes, laddered collateral policies, forbearance protocols with sunset clauses.
- $\downarrow \Lambda_{\mathrm{I}\to\mathrm{A}\to\mathrm{C}}$: cadence-locked communications, transparent metrics, SLA-based dispute handling to avoid polarization spirals.

Buffer design principles.

1. **Time-to-build**: prefer instruments with short activation lags when alerts fire.
2. **Cost per unit of risk reduction**: rank levers by ΔB /cost and the elasticity of CRP to B.
3. **Targeting**: segment buffers by the **lead surplus type** causing max SSI.
4. **Fail-safe**: apply hard caps (e.g., leverage ceilings) during BLACK-HOLE conditions regardless of cost.

We parameterize effects as

$$\Delta C^{(\tau)} = \alpha_\tau u_\tau, \qquad \Delta B = \beta^\top u, \qquad \Delta\Lambda_{\tau\to\tau'} = -\gamma_{\tau\to\tau'} u_{\tau\to\tau'},$$

which plugs directly into the control routines (Section 11.3).

11.3 Multi-Objective Control: Trade-offs and Pareto Frontiers

Policy is cast as **constrained, multi-objective control** over a rolling horizon. The **positive** layer supplies dynamics and indicators; the **normative** layer selects controls.

Objectives.
$\mathbf{J}(u) = (J_G, J_R, J_E, J_S)$ for growth, resilience, equity, and sustainability. Typical instantiations:
$J_G = \sum g(x_t)$ (throughput), J_R penalizes BH flags and large CRP, J_E rewards distributional fairness (e.g., wage-share floors, Roemer-style exploitation bounds), J_S penalizes resource drawdowns/externalities.

Constraints.

- **Phase-map safety:** $\mathrm{SSE} \le \bar{\tau}_{\mathrm{SSE}}$, $\mathrm{CRP} \le \bar{\tau}_{\mathrm{CRP}}$, $\Xi \le \bar{\tau}_{\Xi}$.

- **Chance constraints:** Pr(BH violation) $\leq \alpha$ from historical shock distributions.
- **Budget/feasibility:** $u \in \mathcal{U}$, rate limits $|\Delta u| \leq \overline{d}$.

Solution concepts.

- **Weighted sum**: tune λ's to reflect current mandate; display how solutions move on the Pareto frontier.
- ε**-constraint**: fix acceptable resilience/equity/sustainability levels and minimize growth loss—useful during crisis management.
- **Lexicographic**: hard-priority regimes (e.g., "never enter BH cube"), then optimize secondary goals.

Decision surface and governance.
Outputs to decision makers include (i) a **Pareto frontier** annotated with policy mixes, (ii) projected **phase-map trajectories**, and (iii) **buffer budgets** consumed by each plan. This turns political preference revelation into a transparent slider choice among efficient points, while keeping the **black-hole avoidance** non-negotiable.

Implementation note. The accompanying minimal package (Python/Julia MRE) already provides indicator computation, BH thresholding with persistence/hysteresis, and a toy MPC loop. To operationalize this section, replace the toy state with your sectoral panel, link each lever u to empirically estimated $(\alpha_\tau, \beta, \gamma)$, and run rolling-horizon optimization under the calibrated thresholds.

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12. Philosophy of Science & Meta-Impact

12.1 Testability, Demarcation, and What Becomes Decidable

This framework meets **Popperian demarcation** by stating refutable claims with measurable objects. Three families of tests anchor demarcation:

- **Multi-attractor tests.** If the reduced map $z_{t+1} = F(z_t)$ is S-shaped with a buffer kink, bi-stability follows (Theorem 7.1). Empirically, this implies (i) **kinks** in the response of profits/valuations/throughput to SSI and (ii) **bimodality** or regime-switching in residual dynamics. Absence of kinks and regime evidence across admissible specifications rejects the claim.
- **Limiting-case tests.** When cross-type conversion and buffers are suppressed, the model predicts a **single historical path** (Theorem 7.2). This is empirically visible as a unique, monotone trajectory toward constraint sets during episodes where Λ and B are exogenously muted (e.g., administratively frozen finance).
- **Early-warning tests.** A persistent BH flag must **raise the conditional probability of tail events** within horizon H by a positive margin (Theorem 7.5.3). Failure across threshold grids falsifies the mapping from indicators to dynamics.

Beyond demarcation, several long-disputed questions become **decidable**:

1. **"Historical necessity?"** Decides as a **parameter-region** question: do data support the degenerate conditions that collapse basins into a single path?
2. **"Finance vs. real primacy?"** Decides by comparing **conversion elasticities** $\Lambda_{M\to F}$ and $\Lambda_{F\to I}$ to alternatives; larger elasticities identify the dominant channel.
3. **"Crisis causation taxonomy?"** Decides by attributing the lead pressure to $\max_\tau \mathrm{SSI}^{(\tau)}$ and the subsequent routing via Λ, rather than by narrative labels.
4. **"Policy sequencing?"** Decides on the **Pareto frontier** of multi-objective control under phase-

map constraints, not by a priori ideology.

Under a **Lakatosian** lens, this constitutes a **progressive research programme**: the hard core (surplus as a multi-type driver; cross-domain conversion; thresholds) generates **novel facts**—bifurcation signatures, entropy-based early warnings—that are independently testable and practically useful.

12.2 Scope, Limits, and Ethical Considerations

Scope. The framework targets **meso-to-macro horizons** (quarterly to annual), **cross-domain routing** (material/financial/institutional/attention/cognitive), and **phase-aware governance** (multi-basin geometry and early warnings).

Limits.

- **Measurement sensitivity.** SSE depends on corpus composition and clustering; CRP depends on weights for leverage, gains, and buffers. Robustness requires window and source checks, plus out-of-sample validation.
- **Partial observability and co-identification.** Generation Γ, capacities C, and conversions Λ may be co-linear; identification needs instruments, event studies, or ABM-informed priors.
- **Parameter drift.** Thresholds and gains may shift after structural breaks; periodic recalibration is essential.
- **Model-form risk.** The S-shape is a local approximation; spline/nonparametric fits should be used when data stray from the working domain.
- **Data availability.** Attention/cognitive proxies may be sparse or proprietary; confidence bands must reflect this.

Ethical considerations.

- **Goodhart and gaming.** Publishing thresholds can induce actors to cosmetically **optimize the metric** (e.g., narrative shaping to lower SSE). We mitigate this with (i) **multi-source triangulation**, (ii) manipulation detectors (metric–event divergence), and (iii) audit trails.
- **Fairness in buffers.** Buffer design can unintentionally burden weaker sectors or households. Equity enters explicitly via J_E or hard floors (e.g., wage-share minima, exploitation bounds), ensuring that resilience is not purchased with hidden inequities.
- **Privacy and explainability.** Text-based SSE construction must respect minimization and anonymization; automated decisions must carry **appeal and explanation** channels.
- **Institutional legitimacy.** Because SSE tracks coordination strain, **policy communication** becomes part of the plant: cadence and transparency protocols are integral design choices, not add-ons.

12.3 How the Framework Reduces Long-Running Debates

The meta-impact is to **convert philosophical stalemates into statistical decisions** with clear failure modes:

- The **necessity vs. contingency** dispute narrows to whether data occupy the **degenerate parameter region** that yields a single path. This can be tested and rejected.
- The **value vs. price** quarrel is reframed: material surplus is **measured on a Sraffian base**, then routed across domains; the question becomes how much and how fast conversion occurs, not which theory is “true.”
- The **crisis typology** debate collapses into a **common geometry**: high SSI (where), rising CRP

(how fast), high SSE (coordination failure). Labels give way to trajectories on a phase map.

- The **policy ideology** argument becomes a **Pareto choice** on a transparent frontier with hard safety constraints (black-hole avoidance). Preferences are revealed as weights or ε-bounds rather than polemics.

In short, by standardizing **observables (SSI/CRP/SSE)**, **operators (S1–S4)**, and **governance analytics (phase maps, MPC)**, the framework replaces open-ended argument with **replicable tests, explicit trade-offs, and audited decisions**. Even when rejected or revised, it **saves argumentative resources** by clarifying what evidence would count and how to measure it.

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13. Conclusion & Future Work

Conclusion

This paper developed a **general, testable architecture** for surplus-driven economic dynamics. We generalized surplus from a single accounting residue into a **multi-type driver** spanning material, financial, institutional, attention, and cognitive domains. A **Type-Conversion operator** Λ and domain-specific **absorption capacities** $C^{(\tau)}$ generate rich system behavior when coupled with four flow operators—**Trigger (S1)**, **Feedback (S2)**, **Role/Structure Shift (S3)**, and **Type Conversion (S4)**. Three measurable indicators—**Surplus Saturation (SSI)**, **Collapse Readiness Potential (CRP)**, and **Semantic Saturation Entropy (SSE)**—anchor identification, early warning, and policy control on a **phase map** with explicit **black-hole** avoidance rules.

On the theory side, we proved **sufficient conditions for multi-stability**, showed that the ***ECONOMICS* single-path "necessity"** emerges as a **limiting case** when buffers and conversion are suppressed, and established **consistency bounds** linking BH flags to finite-horizon tail risks. On the empirical side, we specified an operational protocol—**Map → Fire → Render**—to measure indicators, identify thresholds and conversion elasticities, and deploy a **multi-objective MPC** with hard safety constraints. Applications to industrial cycles, financialization episodes, platform economies, and policy stress tests illustrated how **surplus routing and buffers** determine whether shocks are absorbed or transmuted into crises. Finally, by integrating Goodwin, Sraffa, Roemer, DSGE, and ABM within a single geometry, the framework **bridges schools of thought**, turning philosophical disputes into **decidable empirical questions** with auditable trade-offs.

Future Work

(1) Cross-sectional validation and external generalization.
Build multi-country, multi-sector panels to estimate: (i) trigger thresholds via kink/RD designs; (ii) conversion elasticities $\Lambda_{\tau\to\tau'}$ via IV/event studies; (iii) BH predictive power via hazards and ROC/PR. Benchmark portability across institutional regimes and data vintages.

(2) Microfoundations and coarse-graining.
Use ABMs and structural estimation to recover **capacity reaction functions** $C^{(\tau)}(\cdot)$, **feedback gains** $\overline{g}(\cdot)$, and **conversion surfaces** $\Lambda(\cdot)$; aggregate these to the SSI/CRP/SSE layer with error propagation and uncertainty bands.

(3) Semantic measurement advances.
Upgrade SSE with dynamic topic models, semantic graphs, and manipulation detectors (metric–event divergence). Establish multilingual robustness and establish provenance/audit trails for governance use.

(4) Robust, equitable control.
Extend the controller to **distributionally robust MPC** (e.g., Wasserstein balls) with **chance constraints** $\Pr(\mathrm{BH}) \le \alpha$. Incorporate **equity/anti-exploitation floors** (Roemer-style metrics) as hard constraints so that resilience is not purchased via hidden inequities.

(5) Structural-break adaptation.
Develop online learning for **threshold drift** and **gain shifts** after regime changes; combine sequential

testing with re-calibration to maintain out-of-sample validity.

(6) Sustainability and ecological surplus.
Augment the type set with an **ecological surplus** dimension and introduce resource/entropy ledgers; revisit SSE and CRP to account for climate and critical-materials constraints.

(7) Policy sandboxes and scenario libraries.
Codify counterfactual playbooks (e.g., **Redistribution→Innovation** sequencing) as reusable scenario trees; publish outcome distributions on the phase map with buffer budgets and Λ usage, enabling ex-ante accountability.

(8) Open replication and tooling.
Release a full replication package—data dictionaries, code, and dashboard templates—building on the minimal Python/Julia MRE to standardize measurement, alerts, and control workflows across institutions.

Closing.
The promise of **Surplus Dynamics** is not only predictive accuracy but **argument-saving clarity**: it standardizes observables, renders thresholds explicit, and operationalizes governance as constrained control on a transparent phase map. Whether the framework stands or falls is an empirical matter—either way, it **compresses decades of debate into testable claims**, freeing attention and resources for cumulative progress.

Appendix A. Notation and Definitions

A.1 Sets, Indices, and Operators

- $t \in \mathbb{Z}_{\geq 0}$ — time index (periods).
- $i \in \{1, \ldots, N\}$ — sector (or unit) index.
- $\mathcal{T} = \{\mathrm{M}, \mathrm{F}, \mathrm{I}, \mathrm{A}, \mathrm{C}\}$ — surplus types: Material / Financial / Institutional / Attention / Cognitive.
- $\mathbf{1}$ — vector of ones (conformable dimension).
- $\mathrm{vec}(\cdot)$ — column-stacking vectorization.
- $\|\cdot\|_w$ — type-weighted norm (nonnegative weights w).
- $\mathbb{1}\{\cdot\}$ — indicator function.
- $\sigma(z) = 1/(1 + e^{-z})$ — logistic map (used in CRP).

A.2 State, Observables, Controls, Shocks

- $x_t \in \mathcal{X} \subset \mathbb{R}^n$ — system state (macro/financial/institutional): typical components $x_t = (y_t, k_t, h_t, e_t, b_t, q_t, \tau_t, \ldots)$
  where y=output, k=capital, h=wage share, e=employment, b=balance-sheet metrics, q=asset valuations, τ=institutional trust/legitimacy.
- $y_t^{\mathrm{obs}} = h_{\mathrm{meas}}(x_t) + \eta_t$ — observables with measurement noise η_t.
- $u_t \in \mathcal{U}$ — control vector (policy/operational levers).
- $\varepsilon_t \sim \mathcal{E}$ — exogenous shocks (technology, demand, liquidity, regulatory, information).

A.3 Surplus, Capacity, Buffers

- $\sigma_t = \left[\sigma_{i,t}^{(\tau)}\right] \in \mathbb{R}_{\geq 0}^{N \times |\mathcal{T}|}$ — **surplus tensor** (excess pressure by sector & type).
- $C_t = \left[C_{i,t}^{(\tau)}\right] \in \mathbb{R}_{>0}^{N \times |\mathcal{T}|}$ — **absorption capacities** (domain-specific "stomach").
- $B_t \in \mathbb{R}_{\geq 0}$ — **buffers** aggregate (capital/liquidity/inventories/fiscal space/social trust).
- $z_t = w^{\top}\mathrm{vec}(\sigma_t)$ — scalar order parameter (nonnegative weights w) used for reduced dynamics.

A.4 System Maps and Evolution

- **Core dynamics**

$$x_{t+1} = f(x_t, \sigma_t, u_t, \varepsilon_t), \qquad \sigma_{t+1} = \Gamma(x_{t+1}) + \Lambda_t\, \sigma_t - D(x_t, \sigma_t).$$

- $\Gamma: \mathcal{X} \to \mathbb{R}_{\geq 0}^{N \times |\mathcal{T}|}$ — surplus **generation** map.
- $D: \mathcal{X} \times \mathbb{R}^{N \times |\mathcal{T}|} \to \mathbb{R}_{\geq 0}^{N \times |\mathcal{T}|}$ — **absorption/dissipation** map.
- $\Lambda_t \in \mathbb{R}_{\geq 0}^{|\mathcal{T}| \times |\mathcal{T}|}$ — **type-conversion** operator (row-sub-stochastic: row sums ≤ 1).
- R_t — role/structure (bargaining, governance, platform architecture).

A.5 Surplus Flow Operators $S1$–$S4$

- $S1$ **Trigger:** acceleration/leverage when saturation exceeds thresholds

$$\Delta x_{t+1}^{(S1)} = A_1(x_t)\, \mathbb{1}\{\mathrm{SSI}_{i,t}^{(\tau)} \geq \theta_1^{(\tau)}\}.$$

- $S2$ **Feedback:** amplification via profits/prices/throughput

$$g_t = \partial\, m_{t+1} / \partial\, \sigma_t, \quad \Delta x_{t+1}^{(S2)} = A_2(x_t)\, \phi(g_t, \Delta m_t),$$

  with ϕ S-shaped.

- $S3$ **Role/Structure Shift:** discrete evolution of bargaining/governance

$$R_{t+1} = \Phi(R_t; \mathrm{SSI}_t, \mathrm{CRP}_t, u_t), \quad \Delta x_{t+1}^{(S3)} = A_3(x_t)\, \psi(R_{t+1} - R_t).$$

- $S4$ **Type Conversion:** cross-domain routing with losses

$$\sigma_{t+1}^{(\tau')} \supset \sum_{\tau} (\Lambda_t)_{\tau \to \tau'}\, \sigma_t^{(\tau)}\, (1 - \delta_{\tau \to \tau'}), \quad 0 \leq \delta_{\tau \to \tau'} < 1.$$

A.6 Indicators (Definitions & Axioms)

- **Surplus Saturation Index (SSI)** for each (i, τ):

$$\mathrm{SSI}_{i,t}^{(\tau)} = \min\left\{1,\ \frac{\sigma_{i,t}^{(\tau)}}{C_{i,t}^{(\tau)} + \epsilon}\right\} \in [0, 1].$$

Axioms: monotone in σ, decreasing in C, scale-free, bounded.

- **Collapse Readiness Potential (CRP):**

$$\mathrm{CRP}_t = \sigma\left(\alpha L_t + \beta\, \overline{g}_t - \gamma B_t\right) \in [0, 1],$$

where L_t=leverage/maturity risk, $\bar{g}_t$=aggregated feedback gains, B_t=buffers.
Axioms: $\uparrow L, \uparrow \bar{g} \Rightarrow \uparrow \mathrm{CRP}$; $\uparrow B \Rightarrow \downarrow \mathrm{CRP}$.

- **Semantic Saturation Entropy (SSE):**

$$\mathrm{SSE}_t = \frac{-\sum_{k=1}^{K_t} \pi_{k,t} \log \pi_{k,t}}{\log K_t} \in [0,1],$$

with $\Pi_t = (\pi_{1,t}, \ldots, \pi_{K_t,t})$ narrative-cluster shares.
Axioms: rises with polarization/fragmentation; normalized for varying K_t.

A.7 Thresholds, Flags, and Phase Map

- Up-thresholds $(\tau_{\mathrm{SSE}}, \tau_{\mathrm{CRP}}, \tau_{\Xi}) \in (0,1)^3$; down-thresholds $(\tau_{\overline{\mathrm{SSE}}}, \tau_{\overline{\mathrm{CRP}}}, \tau_{\overline{\Xi}})$ for hysteresis; **dwell time** $d_{\min} \in \mathbb{N}$.

- **Composite index:** $\Xi_t = w_1\,\mathrm{SSE}_t + w_2\,\mathrm{CRP}_t + w_3 \max_\tau \mathrm{SSI}_t^{(\tau)}$ with $w_i \geq 0,\ \sum w_i = 1$.

- **BH flag** at t if either
(BH-1) SSE, CRP $\geq$ thresholds for $d_{\min}$ periods; or
(BH-2) $\Xi \geq \tau_{\Xi}$ for $d_{\min}$ periods.

- **Phase map:** coordinates $\left(\max_\tau \mathrm{SSI}^{(\tau)}, \mathrm{CRP}, \mathrm{SSE}\right) \in [0,1]^3$ partitioned into **SAFE/CRITICAL/ BLACK-HOLE** regions.

A.8 Geometry, Stability, and Invariants

- **Reduced map:** $z_{t+1} = F(z_t) = S(z_t) - B(z_t) + c$ with S S-shaped, B nondecreasing (possibly kinked).

- **Attractors & basins:** fixed points/sets with basins $\mathcal{B}(\cdot)$; unstable manifold W^u forms basin boundaries.

- **Mass non-creation under pure conversion:** if $\Gamma = D = 0$ and Λ_t row-sub-stochastic, then $\mathbf{1}^\top \sigma_{t+1} \leq \mathbf{1}^\top \sigma_t$.

- **Accounting identity (general):**

$$\mathbf{1}^\top \sigma_{t+1} - \mathbf{1}^\top \sigma_t = \mathbf{1}^\top \Gamma(x_{t+1}) - \mathbf{1}^\top D(x_t, \sigma_t) - \mathbf{1}^\top \Delta_{\mathrm{loss},t}.$$

A.9 Policy Layer: Objectives and Constraints

- **Objectives (per horizon T)**:

$$\mathbf{J}(u) = \left(J_G, J_R, J_E, J_S\right) = \sum_{t=0}^{T} \left(g(x_t),\ r(x_t),\ e(x_t),\ s(x_t)\right),$$

for Growth / Resilience / Equity / Sustainability.

- **Safety constraints:** $\mathrm{SSE} \leq \bar{\tau}_{\mathrm{SSE}},\ \mathrm{CRP} \leq \bar{\tau}_{\mathrm{CRP}},\ \Xi \leq \bar{\tau}_{\Xi}$; chance constraints $\Pr(\mathrm{BH}) \leq \alpha$.

- **Lever effects (linearized):** $\Delta C^{(\tau)} = \alpha_\tau u_\tau,\ \Delta B = \beta^\top u,\ \Delta\Lambda_{\tau\to\tau'} = -\gamma_{\tau\to\tau'} u_{\tau\to\tau'}$.

A.10 Data-to-Symbol Mapping (Proxy Cheat-Sheet)

- **Material** $\sigma^{(M)}$**:** price–cost margins, utilization, inventory cover; $C^{(M)}$: utilization ceilings, inventory policy, market depth.
- **Financial** $\sigma^{(F)}$**:** leverage (DSR, debt/EBITDA), spreads/haircuts, duration mismatch; $C^{(F)}$: capital & liquidity ratios.
- **Institutional** $\sigma^{(I)}$**:** rule velocity, enforcement backlog, conflict indices; $C^{(I)}$: staffing, automation, trust.
- **Attention** $\sigma^{(A)}$**:** ad load, push intensity, dwell saturation, churn; $C^{(A)}$: audience time, subscription depth.
- **Cognitive** $\sigma^{(C)}$**:** decision latency, error/override rate, burnout/turnover; $C^{(C)}$: automation, specialization, slack.
- **Buffers** B**:** capital/liquidity/inventories/fiscal space/trust composites.
- **SSE inputs** Π_t**:** clustered text shares from news, policy docs, earnings calls, platform data.

A.11 Normalization and Conventions

- **Scaling:** all indicators mapped to $[0, 1]$; winsorize raw proxies (1st–99th pct), z-score by sector; apply logistic/rank scaling before capping in SSI.
- **Thresholds:** select $(\tau_{\cdot}, d_{\min})$ on validation windows via ROC/PR F_β trade-offs; use hysteresis $(\tau_{\cdot}^{-})$ to avoid alert ping-pong.
- **Units:** monetary series deflated; rates in percentage points; entropies unitless; capacities in consistent quantity/time units; buffers in standardized scores.
- **Estimation:** cluster-robust errors; unit/time fixed effects for panels; IV/event-study for Λ; local projections/splines for S-shapes.

Appendix B. Proofs of Propositions and Theorems

Notation follows Appendix A. Throughout, continuity means at least C^1 unless stated; expectations are conditional on the natural filtration $\{\mathcal{F}_t\}$.

B.0 Preliminaries (lemmas & tools)

Lemma B.0.1 (Logistic slope bound).
For $S(z) = \dfrac{L}{1 + e^{-k(z - z_0)}}$ with $L, k > 0$, $S'(z) = \dfrac{kL}{4}\ \mathrm{sech}^2\left(\frac{k}{2}(z - z_0)\right)$ and $\max_z S'(z) = \dfrac{kL}{4}$.

Lemma B.0.2 (Piecewise-affine buffer).
Let $B(z) = b_0 + b_1 z + \eta\, \mathbb{1}\{z \geq \theta\}$ with $b_1 \in (0, 1)$, $\eta \geq 0$. Then B is nondecreasing; the left/right derivatives are b_1 except at $z = \theta$ where a jump of size η occurs.

Lemma B.0.3 (Fixed point stability in 1-D).
For $z_{t+1} = F(z_t)$, a fixed point $z^\star$ with $|F'(z^\star)| < 1$ is locally asymptotically stable; if $F'(z^\star) > 1$ it is (locally) unstable.

Lemma B.0.4 (Block-Jacobian lifting).
Let the full Jacobian at an equilibrium be

$$J = \begin{pmatrix} A & B \\ C & d \end{pmatrix}, \quad d = F'(z^\star),$$

where $z = w^\top \mathrm{vec}(\sigma)$ and $\rho(A) < 1$. Then if $|d| < 1$ and $\|B\|, \|C\|$ are sufficiently small (Gershgorin/ continuity), the spectral radius $\rho(J) < 1$ and the equilibrium is locally asymptotically stable for the full system.

Lemma B.0.5 (Freedman's inequality for martingales).
If $\{M_t\}$ is a martingale difference with bounded increments $|M_t| \le M$ and conditional variance process V_t, then for any $x, v > 0$

$$\Pr\left(\sum_{s=1}^{t} M_s \ge x \,\&\, V_t \le v\right) \le \exp\left(-\frac{x^2}{2(v + Mx/3)}\right).$$

B.1 Proof of Theorem 7.1 (Existence of multiple attractors)

Restatement.
Let $z_{t+1} = F(z_t) = S(z_t) - B(z_t) + c$ with logistic S and piecewise-affine B as in Lemmas B.0.1–B.0.2. Suppose $\max_z S'(z) - b_1 > 1$ (equivalently $kL/4 - b_1 > 1$) and choose (η, θ) so that the buffer kink lies near the inflection of S. Then $F(z) = z$ admits exactly three fixed points $z^- < z^0 < z^+$ with $|F'(z^\pm)| < 1$, $F'(z^0) > 1$. With a full-system Jacobian whose transverse block has spectral radius < 1, these lift to two asymptotically stable equilibria with disjoint basins.

Proof.

1. *Three intersections with the identity.*
   Define $G(z) = F(z) - z = S(z) - B(z) + c - z$. As $z \to -\infty$, $S(z) \to 0$ and $G(z) \sim -(b_1 + 1)z - (b_0 + \eta \cdot 0 - c) \to +\infty$ (since $-(b_1 + 1)z \to +\infty$).
   As $z \to +\infty$, $S(z) \to L$ and $G(z) \sim -(b_1 + 1)z + (L - b_0 - \eta - c) \to -\infty$.
   By continuity, at least one root exists. The derivative is

$$G'(z) = S'(z) - b_1 - 1,$$

so at the logistic peak, $G'(z_0) = kL/4 - b_1 - 1 > 0$ by assumption, while in the tails $S'(z) \to 0$ and $G'(z) \to -b_1 - 1 < 0$. Hence G transitions from decreasing to increasing (near z_0) back to decreasing, implying **three** simple roots for a generic vertical shift c (by the intermediate value theorem and strict monotonic segments). Choosing η, θ aligns the kink to avoid tangency (genericity).

2. *Stability in 1-D.*
   At the outer roots $z^\pm$, $S'(z^\pm)$ is small (logistic tails), so $F'(z^\pm) = S'(z^\pm) - b_1 \in (-b_1, b_1) \subset (-1, 1)$ using $b_1 \in (0, 1)$.
   At the middle root z^0 near the logistic peak, $F'(z^0) = S'(z^0) - b_1 > 1$ (strictly), giving instability. Lemma B.0.3 applies.

3. *Lifting to the full system.*
   Let the full Jacobian at $(x^\pm, \sigma^\pm)$ be block-partitioned with scalar bottom-right $d = F'(z^\pm) \in (-1, 1)$ and transverse block A capturing directions orthogonal to z. Under the standing assumption $\rho(A) < 1$ (transverse contraction; achievable by design of buffers/capacities near equilibria), continuity ensures $\rho(J) < 1$ for small enough $\|B\|, \|C\|$. Therefore both outer fixed points are locally asymptotically stable; the middle one has $|d| > 1$ and is unstable. Different basins follow from the standard 1-D picture and the stable manifold theorem. ■

B.2 Proof of Theorem 7.2 (*ECONOMICS* single-path as a limiting case)

Restatement.
With a single type $\mathcal{T} = \{\mathrm{M}\}$, no buffers and no conversion ($B \equiv 0,\ \Lambda \equiv I$), take $S(z) = az + b$ with $a > 1$. Then the map $z_{t+1} = az_t + b$ has a unique monotone trajectory tending to an absorbing crisis boundary or to the unique unstable threshold $z^\star = \frac{b}{1-a}$; there are no competing basins.

Proof.
The closed form is $z_t = a^t z_0 + b\sum_{k=0}^{t-1} a^k = a^t z_0 + b\ \frac{a^t - 1}{a-1}$, strictly increasing in t for $a > 1, b \geq 0$. The fixed point $z^\star = \frac{b}{1-a}$ (if considered) has derivative $F'(z^\star) = a > 1$ and is unstable. Define a crisis set $\partial\Omega = \{z \geq z_{\mathrm{crit}}\}$ (e.g., realization/solvency boundary). Then the hitting time $T = \inf\{t:\ z_t \geq z_{\mathrm{crit}}\}$ is finite (or finite in expectation if $b = 0$ and the boundary is at infinity). Since the map is strictly increasing and 1-D, there cannot exist two distinct attracting fixed points or cycles. Hence a single path. ■

B.3 Proof of Proposition 7.3.1 (Saddle-node & cusp)

Saddle-node.
Let $G(z;\mu) = F(z;\mu) - z$. At $(z^\star, \mu^\star)$ suppose $G = 0$, $\partial_z G = 0$, $\partial_{zz} G \neq 0$, $\partial_\mu G \neq 0$. By the saddle-node bifurcation theorem for 1-D maps, as μ crosses $\mu^\star$ a pair of fixed points is created/annihilated. In our setting, the logistic slope controls $\partial_z G$ and the buffer kink alters G locally; choosing $kL/4 - b_1$ just above 1 and placing θ near z_0 gives the nondegeneracy.

Cusp.
With two parameters, say (k, b_1), impose $G = 0$, $\partial_z G = 0$, $\partial_{zz} G = 0$, but $\partial_{zzz} G \neq 0$ and the 2×2 Jacobian of $(\partial_\mu G, \partial_{\mu'} G)$ w.r.t. (μ, μ') has full rank. Then a cusp organizes regions of mono-/bi-stability. The logistic third derivative at the inflection is nonzero; the buffer kink shifts intercept/slope to satisfy rank conditions generically. ■

B.4 Proof of Proposition 7.3.2 (Neimark–Sacker)

Consider the full map in a 2-D center subspace near an equilibrium. If the Jacobian has a complex-conjugate pair $\lambda(\mu), \overline{\lambda(\mu)}$ with $|\lambda(\mu^\star)| = 1$, $\frac{d}{d\mu}|\lambda|\Big|_{\mu^\star} \neq 0$, and the usual nonresonance/first Lyapunov coefficient conditions hold, then by the discrete Hopf (Neimark–Sacker) theorem, a closed invariant curve is born. In our model, S-shaped feedback affects the center-manifold coefficients; buffer and conversion parameters (b_1, Λ) can push $|\lambda|$ across 1. ■

B.5 Proof of Proposition 7.3.3 (Path dependence & hysteresis)

Let $\Phi: \mathcal{X} \rightrightarrows \mathcal{X}$ be the set-valued map induced by up/down thresholds (hysteresis) and dwell-time (persistence). Φ is outer semicontinuous with compact, nonempty values. The unstable manifold W^u of the middle fixed point z^0 partitions the phase space into two open basins $\mathcal{B}^{\pm}$. A shock moving the state across W^u leads to convergence toward the other attractor. With hysteresis, the *return* correspondence requires meeting lower thresholds $\tau^- < \tau$ for $d_{\min}$ periods, which, unless a sufficiently strong counter-shock is applied, is generically not satisfied—yielding path dependence. ■

B.6 Proof of Proposition 7.4.1 (Mass non-creation) and 7.4.2 (Accounting identity)

7.4.1.
Under pure conversion $\Gamma = D = 0$ and row-sub-stochastic Λ_t,

$$\mathbf{1}^\top \sigma_{t+1} = \mathbf{1}^\top \Lambda_t \, \sigma_t \le \mathbf{1}^\top \sigma_t,$$

since each row sum is ≤ 1 and entries are nonnegative.

7.4.2.
From $\sigma_{t+1} = \Gamma(x_{t+1}) + \Lambda_t \, \sigma_t - D(x_t, \sigma_t) - \Delta_{\text{loss},t}$, summing over all indices and using $\mathbf{1}^\top \Lambda_t \, \sigma_t \le \mathbf{1}^\top \sigma_t$ gives

$$\mathbf{1}^\top \sigma_{t+1} - \mathbf{1}^\top \sigma_t = \mathbf{1}^\top \Gamma(x_{t+1}) - \mathbf{1}^\top D(x_t, \sigma_t) - \mathbf{1}^\top \Delta_{\text{loss},t}.$$

This is the stated identity. ■

B.7 Proof of Theorem 7.5.1 (Threshold identification via kink)

Let M_{it} be an outcome (profits/valuation/throughput) whose reduced form respects S1:

$$M_{it} = a(s_{it}) + \delta \, \mathbb{1}\{s_{it} \ge \theta_1\} + \nu_{it}, \quad s_{it} \equiv \text{SSI}_{it}^{(\tau)},$$

with $a(\cdot)$ continuously differentiable and ν_{it} mean zero given s_{it}. Let an instrument Z_{it} shift **generation** for type τ but not **capacity** nor ν_{it} (exclusion), and be relevant for s_{it} (monotonicity). Then the conditional expectation $m(s) \equiv \mathbb{E}[M_{it} \mid s_{it} = s, Z_{it}]$ satisfies:

- continuity of $m(s)$ at $s \ne \theta_1$,
- a slope jump at $s = \theta_1$:

$$\lim_{h \downarrow 0} m'(\theta_1 + h) - \lim_{h \downarrow 0} m'(\theta_1 - h) = \delta \cdot a_s(\theta_1) \ne 0,$$

by the chain rule and the activation of the indicator. Thus θ_1 is identified as the unique kink of $m(s)$. In practice, estimate with regression–kink or piecewise-linear IV, instrumenting s with Z and allowing different slopes on each side; the kink location is point-identified under standard regularity and a unique kink assumption. ■

B.8 Proof of Theorem 7.5.2 (Conversion elasticity identification)

Consider the panel equation for two types τ, τ':

$$\sigma_{it}^{(\tau')} = \lambda_{\tau \to \tau'} \, \sigma_{it}^{(\tau)} + X_{it}\beta + \alpha_i + \gamma_t + u_{it},$$

where X_{it} are controls, α_i unit-FE, γ_t time-FE. Let $Z_{it}^{(\tau)}$ be instruments that shift $\sigma^{(\tau)}$ (e.g., policy or tech shocks specific to τ) but are excluded from $\sigma^{(\tau')}$ **except** through $\sigma^{(\tau)}$ contemporaneously. Relevance: $\text{Cov}(Z^{(\tau)}, \sigma^{(\tau)} \mid \alpha_i, \gamma_t) \ne 0$. Exclusion: $\mathbb{E}[Z^{(\tau)} u_{it} \mid \alpha_i, \gamma_t] = 0$.

Under rank and exclusion, the 2SLS estimand consistently recovers

$$\lambda_{\tau \to \tau'} = \frac{\text{Cov}\big(Z^{(\tau)}, \sigma^{(\tau')} \mid \alpha_i, \gamma_t, X\big)}{\text{Cov}\big(Z^{(\tau)}, \sigma^{(\tau)} \mid \alpha_i, \gamma_t, X\big)}.$$

Dynamic lags can be added to purge reverse timing. Thus conversion elasticity is identified locally. ■

B.9 Proof of Theorem 7.5.3 (Falsifiability of BH rule via hitting-time bounds)

Restatement (linking to Theorem 3).
Define the BH-region $\mathcal{H} = \{\mathrm{SSE} \geq \tau_{\mathrm{SSE}},\ \mathrm{CRP} \geq \tau_{\mathrm{CRP}}\}$. Assume:
(H1) *Negative drift*: there exists a Lyapunov function

$$V_t = \lambda_1(\mathrm{SSE}_t - \tau_{\mathrm{SSE}})_+ + \lambda_2(\mathrm{CRP}_t - \tau_{\mathrm{CRP}})_+ + \lambda_3 \max_{\tau} \mathrm{SSI}_t^{(\tau)}$$

with $\mathbb{E}[V_{t+1} - V_t \mid \mathcal{F}_t] \leq -\kappa < 0$ on $\mathcal{H} \setminus \mathcal{C}$, where $\mathcal{C}$ is a crisis/absorbing set.
(H2) *Bounded increments / sub-Gaussian noise*: $|V_{t+1} - V_t| \leq M$ and conditional sub-Gaussianity with proxy σ^2.
(H3) *Persistence*: the BH-flag requires membership in $\mathcal{H}$ for $d_{\min}$ periods; the probability of exiting $\mathcal{H}$ before hitting $\mathcal{C}$ is $\leq \delta \ll 1$.

Claim. There exist H and $\rho > 0$ such that, conditional on a BH-flag at t, $\Pr(T_{\mathcal{C}} \leq H \mid \mathcal{F}_t) \geq \rho$, with the explicit lower bound

$$\Pr(T_{\mathcal{C}} \leq H) \geq 1 - \exp\left(- \frac{(\kappa H - \Delta V_0)_+^2}{2H\sigma^2 + \frac{2}{3}M(\kappa H - \Delta V_0)_+} \right) - O(\delta),$$

where $\Delta V_0 = \max\{0, V^\dagger - V_t\}$ and $V^\dagger = \inf\{V:\ (x,\sigma) \in \mathcal{C}\}$.

Proof.
Define the martingale differences $M_s = (V_s - \mathbb{E}[V_s \mid \mathcal{F}_{s-1}])$, so

$$V_{t+h} - V_t = \sum_{s=t}^{t+h-1} \left(\mathbb{E}[V_{s+1} - V_s \mid \mathcal{F}_s] + M_{s+1} \right) \leq -\kappa h + \sum_{s=t}^{t+h-1} M_{s+1}.$$

Choose $h = H$. If the noise sum stays below $\kappa H - \Delta V_0$, then $V_{t+H} \geq V^\dagger$ (i.e., the crisis set is hit). By Freedman (Lemma B.0.5) with bounded increments M and variance proxy $H\sigma^2$, we obtain

$$\Pr\left(\sum_{s=t}^{t+H-1} M_{s+1} \geq \kappa H - \Delta V_0 \right) \leq \exp\left(- \frac{(\kappa H - \Delta V_0)_+^2}{2H\sigma^2 + \frac{2}{3}M(\kappa H - \Delta V_0)_+} \right).$$

Thus

$$\Pr(T_{\mathcal{C}} \leq H) \geq 1 - \exp(\cdot).$$

Finally, persistence (H3) ensures the process remains in $\mathcal{H}$ with probability $1 - O(\delta)$ until hitting $\mathcal{C}$; multiply the bounds to obtain the stated lower probability. ■

B.10 Remarks on robustness and regularity

- The existence proofs are **generic**: exact values of (η, θ, c) where tangencies occur form measure-zero sets; arbitrarily small perturbations restore transversality.
- Stability lifting requires only **transverse contraction**; any modeling choice that keeps non-z directions dissipative near equilibria suffices.
- The BH hitting-time bound is **conservative**: any stronger tail condition on the noise (e.g., bounded conditional variance) tightens the exponent via Bernstein-type refinements.
- Identification results hold **locally**; dynamic lags, fixed effects, and clustered inference are required

in practice, but do not alter point identification under the stated exclusion and rank conditions.

End of Appendix B.

Appendix C. Algorithms & Pseudocode (Calibration & Early-Warning)

Pseudocode is language-agnostic. Variables and notation follow Appendix A. MRE functions available: `compute_ssi`, `compute_crp`, `compute_sse`, `bh_thresholds`, `mpc_control` (see minimal package).

C.1 Indicator Construction (SSI / CRP / SSE)

Inputs: sectoral/financial/institutional/attention/cognitive proxies; buffers B; text corpus per t.
Outputs: $\widehat{\mathrm{SSI}}_{i,t}^{(\tau)}$, $\widehat{\mathrm{CRP}}_t$, $\widehat{\mathrm{SSE}}_t$.

```
PROC BuildIndicators(data_panel, text_corpus, config):
    FOR each time t:
        # Material
        sigma_M[i]  ← price[i]*y[i] - Σ_j price[j]*A[j,i]*y[i] - wage*labor[i]*y[i] - tax[i]
        C_M[i]      ← f_cap(utilization[i], inventory_cover[i], market_depth[i])
        SSI[i,M]    ← min(1, scale(sigma_M[i]) / (scale(C_M[i]) + ε))

        # Financial
        L           ← leverage_metrics(t)       # DSR, debt/EBITDA, ST funding share
        gain_fin    ← feedback_gain(t)          # return autocorr, Δprofit elasticity
        buffers_F   ← buffer_metrics_F(t)       # capital, LCR/NSFR
        SSI[*,F]    ← normalize(L ⊙ gain_fin, buffers_F)

        # Institutional / Attention / Cognitive (construct σ and C similarly)
        SSI[*,I], SSI[*,A], SSI[*,C] ← domain_specific_blocks(t)

        # CRP (logistic of L, aggregated gains, buffers)
        g_bar       ← aggregate_gains(across domains)
        B           ← aggregate_buffers(t)
        CRP[t]      ← logistic( α*L + β*g_bar - γ*B )

        # SSE from discourse clustering
        E_t         ← embed(text_corpus[t])             # sentence embeddings
        K_t         ← choose_K(HDBSCAN or BIC-KMeans)
        Π_t         ← cluster_proportions(E_t, K_t)
        SSE[t]      ← shannon_entropy(Π_t) / log(K_t)

    RETURN SSI, CRP, SSE
```

Notes: `scale(·)` = winsorize (1–99%), z-score by sector, optional logistic/rank mapping; domain blocks for I/A/C follow §8.1 proxies.

C.2 Early-Warning: Thresholding with Persistence & Hysteresis

Inputs: time series {SSI, CRP, SSE}, thresholds $(\tau_{\cdot}, \tau_{\cdot}^{-})$, dwell $d_{\min}$, weights w.
Outputs: BH flags, region labels, composite Ξ.

```
PROC BHDetector(SSI, CRP, SSE, up, down, dwell, w=(0.4,0.4,0.2)):
    Xi[t]     ← w1*SSE[t] + w2*CRP[t] + w3*max_τ SSI[t,τ]
    FOR t in 1..T:
        cond1 ← ALL s∈[t, t+dwell-1]: SSE[s] ≥ up.SSE AND CRP[s] ≥ up.CRP
```

```
        cond2 ← ALL s∈[t, t+dwell-1]: Xi[s]  ≥ up.XI
        IF cond1 OR cond2: flag[t] ← 1; region[t] ← BLACKHOLE
        ELSE IF (SSE[t] ≥ up.SSE OR CRP[t] ≥ up.CRP OR Xi[t] ≥ up.XI):
            region[t] ← CRITICAL
        ELSE IF (SSE[t] ≤ down.SSE AND CRP[t] ≤ down.CRP AND Xi[t] ≤ down.XI):
            region[t] ← SAFE
        ELSE:
            region[t] ← region[t-1] (default SAFE at t=1)
    RETURN flag, region, Xi
```

C.3 Threshold Selection via ROC/PR Optimization

Goal: choose (\tau_{\mathrm{SSE}},\tau_{\mathrm{CRP}},\tau_{\Xi}, d_{\min}, w) to optimize a recall-weighted F_β with false-alarm cap.
Inputs: historical BH outcomes $Y_{t:t+H}$ (tail events), grid $\mathcal{G}$.
Output: selected thresholds/weights.

```
PROC CalibrateThresholds(SSI, CRP, SSE, Outcomes, gridG, horizon H, β, α_max):
    best_score ← -∞; best_params ← None
    FOR (τ_SSE, τ_CRP, τ_XI, d, w) IN gridG:
        flags, _, _ ← BHDetector(SSI, CRP, SSE, up=(τ_SSE,τ_CRP,τ_XI), down=hysteresis(·), dwell=d, w=w)
        # Label positives if any flag in [t, t+H]
        Yhat[t]   ← MAX flags[t : t+H]
        ROC, PR   ← compute_curves(Yhat, Outcomes)
        Fβ        ← (1+β^2)*Precision*Recall / (β^2*Precision + Recall) at operating point
        FAR       ← false_alarm_rate(Yhat, Outcomes)
        IF FAR ≤ α_max AND Fβ > best_score:
            best_score ← Fβ; best_params ← (τ_SSE, τ_CRP, τ_XI, d, w)
    RETURN best_params
```

C.4 Trigger Threshold Identification (Kink / RD-in-Slope)

Goal: identify $\theta_1^{(\tau)}$ where S1 activates.
Inputs: outcome M_{it}, $\mathrm{SSI}_{it}^{(\tau)}$, instrument Z_{it}.
Output: $\hat{\theta}_1^{(\tau)}$, slope jump, CI.

```
PROC KinkIV(M, SSIτ, Z, FE=(unit,time), bandwidth ℓ):
    # 1) First stage: instrument SSIτ by Z (within FE)
    ŝ_it ← IV_fit(SSIτ_it ~ Z_it + FE)

    # 2) Grid-search kink location θ over [q5%, q95%] of ŝ
    FOR θ in linspace(...):
        # Piecewise linear with different slopes left/right of θ
        fitθ ← IV_fit(M_it ~ (ŝ_it - θ)_+ + (ŝ_it - θ)_- + controls + FE, inst=(Z_it - θ)_± )
        score(θ) ← minimized SSR or info criterion
    θ^ ← argmin_θ score(θ)
    # 3) Report slope jump and robust CIs
    Δslope ← β_right(θ^) - β_left(θ^)
    CI     ← cluster_robust_CI(fit_{θ^})
    RETURN θ^, Δslope, CI
```

C.5 Conversion Elasticities $\Lambda_{\tau\to\tau'}$ (Panel IV / Event Study)

Goal: estimate routing strength from type τ to τ'.
Inputs: $\sigma_{it}^{(\tau)}$, $\sigma_{it+h}^{(\tau')}$, instruments $Z_{it}^{(\tau)}$.
Output: $\hat{\lambda}_{\tau\to\tau'}(h)$, dynamic profile.

```
PROC PanelIV_Lambda(sigmaτ, sigmaτp, Zτ, FE=(unit,time), lags L, leads H):
    FOR h in 0..H:
        # Local projections with IV
```

```
        y_it ← sigmaτp_{i,t+h}
        x_it ← sigmaτ_{i,t}
        model: y_it = λ_h * x_it + Γ(Lags/controls) + α_i + γ_t + u_it
        λ̂_h  ← 2SLS(y_it ~ x_it + FE, instruments = Zτ_it )
        store λ̂_h and SE_h
    RETURN { (h, λ̂_h, SE_h) }
```

Event-study variant: replace $Z^{(\tau)}$ by exogenous shock dummies and include dynamic dummies $\mathbb{1}\{t-k\}$.

C.6 Multi-Objective MPC with Phase-Map Safety

Goal: choose u_t to optimize (J_G, J_R, J_E, J_S) under SSE, CRP, Ξ constraints.
Inputs: model (f, Γ, Λ, D), current (x_t, σ_t), horizon H, bounds, thresholds.
Output: u_t, predicted path.

```
PROC MPC_SurplusDynamics(state_t, model, H, weights λ, safety τ̄, bounds):
    INIT U ← feasible_guess()
    REPEAT until convergence:
        x ← state_t; σ ← σ_t; J ← 0; violation ← 0
        FOR h in 0..H-1:
            x, σ ← rollout_one_step(x, σ, U[h], model)
            SSI, CRP, SSE ← indicators(x, σ)
            Xi ← w1*SSE + w2*CRP + w3*max_τ SSI
            J  ← J + λ_G*(1 - growth(x)) + λ_R*BHpen(CRP,SSE,Xi)
                   + λ_E*inequality_penalty(x) + λ_S*sustain_cost(x)
            IF SSE>τ̄_SSE OR CRP>τ̄_CRP OR Xi>τ̄_XI: violation += bigM
        J ← J + violation
        U ← update_controls(U, grad(J), bounds, rate_limits)
    RETURN U[0], predicted_trajectory
```

Notes: `update_controls` can be SQP/Adam/CMA-ES; for lexicographic safety, put infinite penalty on violations.

C.7 Scenario Trees (Counterfactual / Stress Testing)

Goal: evaluate branches over shocks and policies; prune dominated paths.
Inputs: root state, shock set $\mathcal{S}$, policy menus $\mathcal{U}$, depth H.
Output: tree with node metrics & Pareto set.

```
PROC ScenarioTree(root, shocks S, policies U, depth H):
    node0 ← root
    FOR level = 1..H:
        FOR each node in level-1:
            FOR s in S, u in U:
                child ← simulate_one_step(node.state, s, u)
                compute indicators, BH flags, objectives at child
                attach child to tree
        prune dominated children by (J_G,J_R,J_E,J_S) and safety
    RETURN tree_summary(frontier nodes, phase-map paths, buffer budgets)
```

C.8 Online Recalibration (Threshold/Gain Drift)

Goal: detect structural drift; update $\tau_\cdot, \overline{g}, \Lambda$.
Inputs: rolling window W, change-point tests.
Output: updated parameters with audit logs.

```
PROC OnlineRecalibration(stream, window W, drift_tests):
    buffer ← deque(max=W)
    FOR each new period t:
        buffer.push(observations_t)
        IF buffer.full():
```

```
            # 1) Drift in gains and thresholds
            test1 ← CUSUM_kink(buffer.SSI→M)
            test2 ← PageHinkley(buffer.CRPs vs outcomes)
            IF test1 or test2 == drift:
                (τ, d, w) ← CalibrateThresholds(buffer, gridG, H, β, α_max)
            # 2) Re-estimate Λ with rolling IV
            Λ̂ ← PanelIV_Lambda(buffer)
            log_changes_to_registry(τ, d, w, Λ̂)
    RETURN registry
```

C.9 SSE Manipulation Detection (Metric–Event Divergence)

Goal: detect gaming (e.g., narrative engineering) when SSE falls but conflict/events rise.
Inputs: SSE, event counts E_t (e.g., disputes, recalls), sentiment/volatility V_t.
Output: manipulation flag.

```
PROC DetectManipulation(SSE, Events, Volatility, lookback L):
    ΔSSE  ← difference(SSE, L)
    ΔEvt  ← difference(Events, L)
    ΔVol  ← difference(Volatility, L)
    score ← w1*(-ΔSSE) + w2*(ΔEvt) + w3*(ΔVol)
    IF score ≥ κ (calibrated on history):
        return FLAG_MANIPULATION
    ELSE:
        return OK
```

C.10 Robust Control (Chance Constraints / Distributionally Robust)

Goal: enforce Pr(BH violation) $\leq \alpha$ under shock uncertainty; or worst-case within Wasserstein ball $\mathbb{B}_\epsilon$.
Inputs: shock samples $\{\varepsilon^{(m)}\}$ or ambiguity set radius ϵ.

```
PROC RobustMPC(state_t, model, H, safety τ¯, α, shocks {ε^m}):
    INIT U
    REPEAT:
        violations ← 0
        FOR each scenario m:
            x, σ ← rollout under shocks ε^m and U
            count_m ← Σ_{h} 1{ SSE>τ¯SSE or CRP>τ¯CRP or Xi>τ¯XI }
            violations += 1{ count_m > 0 }
        chance ← violations / M
        J ← nominal_objective + ρ*max(0, chance - α)
        U ← update_controls(U, grad(J))
    RETURN U[0]
```

Distributionally robust: replace empirical chance with worst-case risk via dual representation over $\mathbb{B}_\epsilon$.

C.11 Early-Warning Consistency Check (Hitting-Time Bound)

Goal: empirical check of Theorem 7.5.3—BH flags must predict tail risks within H.
Inputs: BH flags, outcomes Y, horizon H.
Output: lower-bound verification; calibration of κ, M, σ.

```
PROC ConsistencyCheck(BH_flags, Outcomes, H):
    # Empirical lower bound: compare P(Y_{t:t+H}=1 | BH=1) vs BH=0
    p1 ← mean( Outcomes within H after BH=1 )
    p0 ← mean( Outcomes within H after BH=0 )
    Δ  ← p1 - p0
    # Nonparametric CI via block bootstrap
    CI ← bootstrap_CI(Δ)
```

```
    RETURN Δ, CI, assert(Δ > 0 with chosen confidence)
```

C.12 Data Pipeline & Reproducibility Skeleton

```
PIPELINE SurplusDynamics:
    ingest → clean/winsorize/scale → BuildIndicators
           → (CalibrateThresholds ↔ ConsistencyCheck)
           → (KinkIV, PanelIV_Lambda)
           → BHDetector → Dashboard (phase-map paths)
           → (MPC_SurplusDynamics / RobustMPC)
           → ScenarioTree → Reports (Pareto, buffer budgets)
           → OnlineRecalibration (registry + audit logs)
```

Implementation Notes

- The **MRE (Python/Julia)** already includes `demo.py/demo.jl` with indicators, BH detection, and toy MPC; replace toy state with your panel and plug in C.4–C.5 estimators.
- All modules should log: parameter version, data window, seeds, and validation scores; publish a **model card** per release.

End of Appendix C.

Appendix D. Data Dictionaries & KPI Mappings

D.1 Entities & Join Keys

- **UNIT_ID** (string) — Country×industry or firm code (e.g., `US_SEMICON`, `FIRM_12345`).
- **TIME** (date/period) — Month/quarter/year; aligned to period end.
- **CURR** (string) — Currency code; monetary series are deflated during processing.
- **SRC_VER** (string) — Data source and version; enables multi-version provenance.
- **SCHEMA_VER** (string) — Version of this dictionary/indicator schema.

  Unless noted, tables key on `(UNIT_ID, TIME)` (technical coefficients and text indexes may differ).

D.2 Core Tables (Field Specifications)

D.2.1 `PANEL_UNIT` (unit master)

Field	Type	Example	Description
UNIT_ID	string	`US_SEMICON`	Primary key
LEVEL	enum	`sector-country` / `firm`	Unit granularity
SECTOR_ISIC	string	`C26`	ISIC/NACE code
COUNTRY	string	`US`	ISO-2/3
POP_WEIGHT	float	0.31	Aggregation weight

D.2.2 `REAL_ACCOUNTS`

Field	Type	Unit	Definition/Transform

Field	Type	Unit	Definition/Transform
P	float	price index	Output price (or deflator)
Y	float	real qty	Real output/shipments
A_ij	float	coeff	IO technical coefficient (may live in `IO_BLOCK`)
WAGE	float	currency	Wage bill
LABOR	float	hours	Labor input
TAX_INDIRECT	float	currency	Indirect taxes/fees
UTIL	float	%	Capacity utilization (0–100)
INV_COVER	float	weeks	Inventory cover
MARGIN_PCT	float	%	Price–cost margin

D.2.3 `FIN_ACCOUNTS`

Field	Type	Unit	Definition
LEV_DSR	float	pp	Debt service ratio
LEV_DEBT_EBITDA	float	x	Leverage multiple
ST_FUND_SHARE	float	%	Short-term funding share
SPREAD_CREDIT	float	bp	Credit spread
HAIRCUT_IDX	float	idx	Collateral haircut index
MOMENTUM_RET	float	%	Return momentum (e.g., 12–1)
LCR	float	%	Liquidity coverage ratio
NSFR	float	%	Net stable funding ratio
CAPITAL_RATIO	float	%	Regulatory capital ratio

D.2.4 `INSTITUTIONAL_REGISTRY`

Field	Type	Unit	Definition
RULE_VELOCITY	float	#/q	Rulemaking/notice count
COMPLEXITY_IDX	float	idx	Regulatory complexity
ENF_BACKLOG	float	cases	Enforcement/approval backlog
TRUST_IDX	float	idx	Institutional trust/legitimacy
DISPUTE_RATE	float	#	Dispute/event rate

D.2.5 `ATTENTION_METRICS`

Field	Type	Unit	Definition
AD_LOAD	float	%	Ad load
PUSH_FREQ	float	#/user/day	Push frequency
DWELL_SAT	float	%	Time saturation
CHURN	float	%	Churn rate
SUB_DEPTH	float	count	Subscription/binding depth

D.2.6 `COGNITIVE_OPS`

Field	Type	Unit	Definition
DEC_LATENCY	float	hours	Decision latency
ERROR_RATE	float	%	Error/override rate
OVERRIDE	float	%	Override share
BURNOUT	float	%	Burnout/turnover
AUTOMATION	float	%	Automation penetration

D.2.7 BUFFERS

Field	Type	Unit	Definition
INV_BUFFER	float	weeks	Target inventory buffer
FISCAL_SPACE	float	% GDP	Fiscal space
SAFE_ASSET_SUP	float	idx	Safe-asset supply
TRUST_BUF	float	idx	Social/institutional trust
LIQUIDITY_BUF	float	%	Cash/liquidity buffer

D.2.8 TEXT_CORPUS_INDEX

Field	Type	Unit	Definition
DOC_ID	string	—	Document key
TIME	date	—	Period
UNIT_ID	string	—	Optional mapping to unit
EMBEDDING	vector	—	Sentence/document embedding
SOURCE	enum	news / policy / call	Source type

D.2.9 OUTCOMES (validation/tail events)

Field	Type	Unit	Definition
TAIL_DROP	bin	0/1	Large output/employment drop
CREDIT_EVENT	bin	0/1	Default/downgrade
DRAWDOWN	float	%	Market max drawdown
UNEMP_SPIKE	bin	0/1	Unemployment spike

D.2.10 POLICY_LEVERS

Field	Type	Unit	Effect Map (see D.5)
U_INV_TARGET	float	weeks	$\uparrow C^{(M)}$
U_CAPITAL_BUF	float	pp	$\uparrow C^{(F)}, \uparrow B$
U_LTV_DSTI	float	%	$\downarrow \Lambda_{M \to F}$
U_ENF_STAFF	float	FTE	$\uparrow C^{(I)}$
U_FREQ_CAP	float	%	$\uparrow C^{(A)}$
U_AUTOMATION	float	%	$\uparrow C^{(C)}$
U_RESOLUTION	bin	0/1	$\downarrow \Lambda_{F \to I}$
U_COMMS_CADENCE	float	days	$\downarrow \Lambda_{I \to A \to C}$

D.3 Indicator Construction Mapping (Columns → Formulas)

D.3.1 Material Surplus $\sigma^{(M)}$ (Sraffian base)

$$\sigma_t^{(M)} = P_t \cdot [(I - A)Y_t] - WAGE_t \cdot LABOR_t - TAX_INDIRECT_t\,.$$

- Inputs: `P, Y, A_ij, WAGE, LABOR, TAX_INDIRECT`.
- Normalization: deflate nominal series; winsorize (1st–99th pct); sectoral z-score.

Capacity $C^{(M)}$: `f_cap(UTIL, INV_COVER, market_depth)`
SSI(M): $\min\{1, \text{scale}(\sigma_M)/(\text{scale}(C_M) + \varepsilon)\}$

D.3.2 Financial

- **Pressure (generation):** `sigma_F = f(LEV_DSR, LEV_DEBT_EBITDA, SPREAD_CREDIT, HAIRCUT_IDX, ST_FUND_SHARE, MOMENTUM_RET)`
- **Capacity:** `C_F = g(CAPITAL_RATIO, LCR, NSFR, LIQUIDITY_BUF)`
- **SSI(F):** same cap-and-scale construction as above.

D.3.3 Institutional / Attention / Cognitive

- **Institutional:** `sigma_I = f(RULE_VELOCITY, COMPLEXITY_IDX, ENF_BACKLOG)`; `C_I = g(TRUST_IDX, ENF_STAFF)`
- **Attention:** `sigma_A = f(AD_LOAD, PUSH_FREQ, DWELL_SAT, CHURN)`; `C_A = g(SUB_DEPTH, audience_time)`
- **Cognitive:** `sigma_C = f(DEC_LATENCY, ERROR_RATE, OVERRIDE, BURNOUT)`; `C_C = g(AUTOMATION, specialization, slack)`
- For each type: monotone mapping → scaling → `min(1, ·)` cap.

D.3.4 CRP (Collapse Readiness Potential)

$$\mathrm{CRP}_t = \sigma\left(\alpha\, L_t + \beta\, \overline{g}_t - \gamma\, B_t\right),$$

with

- L_t: composite of `LEV_*` (PCA or weighted).
- $\overline{g}_t$: feedback gains (e.g., `MARGIN_PCT`, profit/throughput elasticities, `MOMENTUM_RET`).
- B_t: buffers composite (`CAPITAL_RATIO, LCR, NSFR, INV_BUFFER, FISCAL_SPACE, TRUST_BUF`).

D.3.5 SSE (Semantic Saturation Entropy)

Pipeline: embeddings → clustering → cluster shares Π_t → entropy

$$\mathrm{SSE}_t = \frac{-\sum_k \pi_{k,t} \log \pi_{k,t}}{\log K_t}.$$

Robustness: vary K_t, window, and source mix.

D.4 KPI Mappings (Dashboard ↔ Actions)

Each KPI specifies **direction** (higher is better/worse), **acceptable range**, **alert thresholds** (with hysteresis and dwell), and an **action card** (primary levers).

D.4.1 Core KPIs

KPI_ID	Name	Formula	Direction	Thresholds (↑ trigger / ↓ exit)	Action Playbook
KPI_SSI_MAX	Max Surplus Saturation	$\max_\tau \mathrm{SSI}^{(\tau)}$	lower-better	0.75 / 0.65	Trace lead type; expand $C^{(\tau)}$; for M: reduce $\Lambda_{M\to F}$
KPI_CRP	Collapse Readiness	CRP	lower-better	0.75 / 0.60	Raise B (capital/ liquidity/ inventories),

KPI_ID	Name	Formula	Direction	Thresholds (↑ trigger / ↓ exit)	Action Playbook
					damp accelerators
KPI_SSE	Semantic Entropy	SSE	lower-better	0.75 / 0.65	Adjust comms cadence; SLA; content mix
KPI_XI	Composite	$w_1 \mathrm{SSE} + w_2 \mathrm{CRP} + w_3 \max \mathrm{SSI}$	lower-better	0.80 / 0.70	Combined package: buffers + throttles
KPI_BH	Black-Hole Flag	per Appendix C.2	binary	dwell = 3	Fail-safe: hard caps, capital gates, cooling-off

D.4.2 Type-Specific KPIs

KPI	Field	Alert if…	Primary Levers
KPI_SSI_M	$\mathrm{SSI}^{(M)}$	≥ 0.75 for 3 periods	`U_INV_TARGET`, `U_LTV_DSTI`
KPI_SSI_F	$\mathrm{SSI}^{(F)}$	≥ 0.75	`U_CAPITAL_BUF`, `U_RESOLUTION`
KPI_SSI_I	$\mathrm{SSI}^{(I)}$	≥ 0.75	`U_ENF_STAFF`, `U_COMMS_CADENCE`
KPI_SSI_A	$\mathrm{SSI}^{(A)}$	≥ 0.80	`U_FREQ_CAP`, content mix
KPI_SSI_C	$\mathrm{SSI}^{(C)}$	≥ 0.80	`U_AUTOMATION`, role design

D.5 Control → Model Parameter Map (Estimable Elasticities)

For each policy lever u, we maintain linearized elasticities (estimated via event studies/IV):

$$\Delta C^{(\tau)} = \alpha_\tau u_\tau, \qquad \Delta B = \beta^\top u, \qquad \Delta \Lambda_{\tau\to\tau'} = -\gamma_{\tau\to\tau'} u_{\tau\to\tau'}.$$

Lever	α_τ (capacity)	β (buffers)	γ (conversion)
U_INV_TARGET	$\alpha_M > 0$	small	$\gamma_{M\to F} > 0$
U_CAPITAL_BUF	$\alpha_F > 0$	large	$\gamma_{M\to F}, \gamma_{F\to I} > 0$
U_LTV_DSTI	—	medium	$\gamma_{M\to F}$ large
U_ENF_STAFF	$\alpha_I > 0$	medium	$\gamma_{I\to A}$
U_FREQ_CAP	$\alpha_A > 0$	—	$\gamma_{I\to A\to C}$
U_AUTOMATION	$\alpha_C > 0$	—	—
U_RESOLUTION	—	medium	$\gamma_{F\to I}$ large
U_COMMS_CADENCE	—	—	$\gamma_{I\to A\to C}$ medium

> Coefficients are calibrated in §8.3 (“Map → Fire → Render”). This table defines the data–model interface for governance tooling.

D.6 Transforms & Quality Control

Common transforms

- Deflate nominal → real (chain or base-year indices).
- Winsorize at 1st–99th percentile.

- Standardize: within-sector z-score; optionally logistic/rank mapping to $[0, 1]$.

Integrity/consistency checks

1. **Mass non-creation** (pure conversion windows): $\Delta \sum_{\tau} \sigma^{(\tau)} \leq 0$.
2. **Monotonicity:** $\partial \mathrm{SSI}/\partial \sigma \geq 0$, $\partial \mathrm{SSI}/\partial C \leq 0$ (spot checks).
3. **Decoupling alert** (SSE manipulation): SSE ↓ while `DISPUTE_RATE` or volatility ↑ → trigger audit.
4. **Threshold robustness:** SSE trend should not flip under alternative K_t/window/source pools.
5. **Missing data:** <5% → impute; ≥5% → downweight unit or mask. Always set `IMPUTE_FLAG`.

D.7 Versioning & Provenance

- **MODEL_CARD:** `{SCHEMA_VER, INDICATOR_VER, THRESHOLD_BUNDLE, CRP_WEIGHTS, SSE_PIPELINE_VER, WINDOW, H, ROC_PR_SCORE, DATE}`
- **DATA_REGISTRY:** source, extraction date, license, hash; each run writes a `RUN_ID`.
- **ALERT_LOG:** `RUN_ID, UNIT_ID, TIME, KPI, VALUE, THRESHOLD, REGION, ACTION_SET, OWNER.`

D.8 Minimal Column Name Map (to MRE code)

- `sigma_M` ← computed from `REAL_ACCOUNTS`
- `C_M` ← from `UTIL, INV_COVER, market_depth`
- `L` ← composite of `LEV_*`
- `g_bar` ← from `MARGIN_PCT, MOMENTUM_RET, Δprofit/Δthroughput`
- `B` ← composite of `CAPITAL_RATIO, LCR, NSFR, INV_BUFFER, FISCAL_SPACE, TRUST_BUF`
- `Π_t` ← cluster shares from `TEXT_CORPUS_INDEX`
- `SSI[:, τ], CRP, SSE` → inputs to `bh_thresholds()` and `mpc_control()`

Note. This appendix defines the **data and KPI interface contract**. Concrete data sources vary by institution, but the fields and transforms are compatible with the calibration (§8) and governance toolkit (§11). If desired, this specification can be exported as a `YAML/JSON` schema for ETL validation and data warehousing.

Appendix E. Robustness Checks and Alternative Specifications

All robustness routines are designed to be executable with the replication package. We denote a **specification bundle** by `SPEC_ID = {INDICATOR_VER, THRESHOLD_BUNDLE, CRP_LINK, SSE_PIPELINE, IV_STACK, H, Dwell, Weights w, Sample window}`. Each reported result must cite its `SPEC_ID`.

E.1 Measurement Robustness (SSI / CRP / SSE)

E.1.1 SSI (Surplus Saturation)

- **Scaling variants:** z-score vs. rank-inverse-normal vs. min–max; report Kendall's τ between SSI series across scalings.
- **Capacity functions $C^{(\tau)}$:** linear index vs. CES $C = (\sum \omega_j c_j^{\rho})^{1/\rho}$ (ρ∈{0.25,1,4}); **report** effect on exceedance share $\Pr(SSI \geq 0.75)$.
- **Cap operator:** hard cap `min(1,·)` vs. soft cap `logit^{-1}(a·)` (a∈{2,4}); **report** kink tests stability.
- **Aggregation across types:** `max_τ` vs. top-q quantile (q=0.8) vs. L-norm (p∈{2,∞}); **report** BH AUROC delta.

E.1.2 CRP (Collapse Readiness Potential)

- **Link function:** logit vs. probit vs. cloglog; **report** calibration (Brier, ECE).
- **Block weights** (α, β, γ)**:**
  (i) PCA factor, (ii) Bayesian hierarchical shrinkage, (iii) equal weights. **Report** sensitivity tornado for AUROC/AUPRC.
- **Buffer set** B**:** baseline vs. excluding fiscal metrics vs. excluding bank ratios; **report** CRP drift and alert count changes.

E.1.3 SSE (Semantic Saturation Entropy)

- **Embedding:** SBERT vs. TF-IDF SVD (k=300) vs. doc2vec;
- **Clustering:** HDBSCAN vs. K-Means (K from BIC elbow) vs. spectral;
- **Normalization:** $H / \log K$ vs. **permanent-topic matched** entropy;
- **Source mix:** news-only vs. policy-only vs. calls-only vs. pooled (reweighted).
  Report: pairwise correlations, slope of SSE→outcomes, and manipulation-divergence test (E.7).

E.2 Identification Robustness (Kink & IV/Event Study)

E.2.1 Trigger kink $\theta_1^{(\tau)}$

- **Estimators:** piecewise-linear IV; quantile regression kink; segmented spline with unknown knot (grid search with HQIC).
- **Bandwidths & windows:** ±{10, 15, 20} percentile of SSI; **McCrary** density test at $\hat{\theta}$.
- **Placebos:** shift θ by ±5pp and re-estimate—slope jump must vanish.
- **Leakage check:** test for discontinuities in predetermined covariates at $\hat{\theta}$.

E.2.2 Conversion elasticities $\Lambda_{\tau \to \tau'}$

- **Weak-IV guard:** report first-stage F (Kleibergen–Paap rk) and use LIML/JIVE when F<10.
- **Over-ID:** Hansen J with instrument stacks (policy + tech shocks); cluster-robust SEs.
- **Temporal validation:** leads/lags event study with pre-trends (-8..-1) jointly = 0.
- **Exogeneity stress:** "drop-one-instrument" jackknife; **report** range of $\hat{\lambda}$.

E.3 Early-Warning Robustness (BH Rule)

- **Threshold grid:** \tau_{\mathrm{SSE}},\tau_{\mathrm{CRP}},\tau_{\Xi}\in\{0.65..0.85\} step 0.02; $d_{\min} \in \{2, 3, 4, 5\}$; w on simplex (coarse grid).
- **OR/AND variants:** BH if (SSE **or** CRP) & dwell vs. (SSE **and** CRP); **report** cost curves under false-alarm budget α.
- **Horizon H:** {4,8,12} quarters; **report** AUROC/AUPRC, Brier, and reliability curves.
- **Cross-time CV: time-slice** folds to avoid look-ahead.
- **Base-rate shifts:** reweight outcomes by period prevalence; **report** PR curves.
- **Uncertainty:** blocked bootstrap CIs for hit-rate uplift Δp and expected hitting-time quantiles.

E.4 Structural Robustness (Model-Form & Regime)

- **Model-free signatures:** Hamilton regime switching / HMM on residuals; **expect** 2-state likelihood improvements when multi-basin holds.
- **Quantile LP/VAR:** IRFs at τ∈{0.25,0.5,0.75} to test amplification asymmetry (S2).
- **Regime invariance:** IRM (Invariant Risk Minimization) across countries/industries—**penalize** spec that gains only in one environment.
- **Counterfactual invariance:** check that policy elasticities (lever→C, B, Λ) are stable across regimes.

E.5 Alternative Specifications (Dynamics & Indices)

- **Reduced map F:** logistic vs. cubic-spline S-shape vs. piecewise-linear; include **delay** $z_{t+1} = S(z_t, z_{t-1}) - B(z_t)$.
- **Composite Ξ:** linear vs. geometric mean vs. max-operator; report sensitivity.
- **Time-varying thresholds:** $\tau_t = \tau_0 + \rho \cdot Z_t$ with Z_t structural-break indicators.
- **State-space indicators:** Kalman/particle filters for latent SSI/CRP/SSE with measurement error; **report** filter-smoother gaps.
- **Networked Λ:** sparse network (Graphical Lasso) vs. low-rank factor mapping; **report** routing centrality shifts.

E.6 Placebo & Null Experiments

- **Permutation:** shuffle BH flags across time within unit; AUROC should drop to ~0.5.
- **Random thresholds:** draw τ from U[0.6,0.85]; alert rate vs. hit rate should match null band.
- **Unit placebo:** apply BH to sectors with no exposure to the shock; require non-significant uplift.
- **Synthetic control:** for treated units with BH alerts, build synthetic doppelgängers; demand significant divergence only post-alert.

E.7 Manipulation/Gaming Detection (SSE)

- **Divergence score:** $S_t = w_1(-\Delta\text{SSE}) + w_2\,\Delta\text{DISPUTE_RATE} + w_3\,\Delta\text{VOL}$.
  Flag if $S_t \geq \kappa$ (calibrated at 95th pct).
- **Change-point:** Bai–Perron multiple breaks on SSE vs. event series; align breaks with known comms interventions.
- **Source triangulation:** recompute SSE excluding the top source; **require** stability of direction/sign.

E.8 Subsamples, Jackknife & Leave-One-Out

- **Subsamples:** pre/post policy regime; advanced vs. emerging; tradable vs. non-tradable.
- **Jackknife units:** leave-one-industry-out; report max deviation in $\hat{\theta}$, $\hat{\lambda}$, AUROC.
- **Source LOO (text):** drop one corpus family at a time; report SSE elasticity stability.

E.9 Uncertainty Quantification & Model Averaging

- **Bootstrap:** block bootstrap (moving blocks) for thresholds and Λ; 1,000 reps.
- **Bayesian:** posterior for (θ_1, Λ) with weakly informative priors; **report** 50/80/95% bands.
- **Stacking:** stack across `SPEC_ID` variants using log predictive density; report **stacked** BH performance.

E.10 External Validity & Transportability

- **Reweighting:** importance weights to match covariate shift $p_{\text{target}}(X)/\,p_{\text{source}}(X)$.
- **Transport tests:** train on region A, test on region B; require non-degraded calibration (ECE ≤ 5pp).
- **Policy portability:** re-estimate lever elasticities in target domain; compare to source via Chow tests.

E.11 Reporting & Governance

- **Spec ledger:** publish a CSV with one row per `SPEC_ID` and columns: metrics (AUROC, AUPRC, Brier, ECE), $\hat{\theta}$ CIs, $\hat{\lambda}$ profiles, alert counts, base rates.
- **Model cards:** for each release, include data windows, excluded sources, drift detections, and any parameter overrides.
- **Pre-registration:** register the primary `SPEC_ID` before out-of-sample evaluation; label all others as exploratory.

E.12 Minimal Routines (link to MRE)

- **Plug-ins:**

- Threshold grid search → `CalibrateThresholds` (App. C.3)
- BH detection → `bh_thresholds`
- MPC safety runs → `mpc_control` / `RobustMPC`
- Kink & IV templates → App. C.4–C.5

Execution order: BuildIndicators → (E.1 variants) → Kink/IV robustness (E.2) → BH grid (E.3) → Structural tests (E.4–E.5) → Placebos (E.6) → Manipulation (E.7) → LOO (E.8) → UQ/Stacking (E.9) → Transport (E.10) → Ledger (E.11).

What to look for (decision rules)

- **Pass** if all primary findings (multi-basin evidence, positive BH uplift, significant $\hat{\lambda}$) hold across: (i) ≥2 SSI scalings, (ii) ≥2 CRP weightings, (iii) ≥2 SSE pipelines, (iv) ≥2 thresholds bundles, with AUROC drop ≤ 0.05 and $\hat{\theta}$, $\hat{\lambda}$ within 95% bands of baseline.
- **Fail** if any single corpus or instrument drives results (jackknife outlier), or if placebos yield AUROC > 0.6.
- **Flag** if manipulation divergence triggers in ≥2 consecutive windows; require audit and potential source reweighting.

End of Appendix E.

Appendix F. Replication Package & Open Problems

F.1 Package Overview & Folder Tree

```
surplus-dynamics-replication/
├── code/
│   ├── python/                # pyproject.toml; src/; scripts/
│   │   ├── src/sd_indicators.py
│   │   ├── src/sd_thresholds.py
│   │   ├── src/sd_kink_iv.py
│   │   ├── src/sd_lambda_iv.py
│   │   ├── src/sd_mpc.py
│   │   └── scripts/{build_indicators,calibrate,bh_detect,run_mpc,make_figs}.py
│   └── julia/
│       ├── Project.toml
│       ├── src/SurplusDynamics.jl
│       └── demos/run_all.jl
├── config/
│   ├── paths.yaml             # data locations
│   ├── indicators.yaml        # proxy → index mapping
│   ├── thresholds.yaml        # τ_SSE, τ_CRP, τ_XI, dwell, weights
│   ├── instruments.yaml       # IV stacks for Λ, kink
│   └── policies.yaml          # lever → (α, β, γ)
├── data/
│   ├── raw/                   # (empty, user-supplied)
│   ├── processed/             # generated parquet/csv
│   └── synthetic/             # demo_panel.parquet; text_sse_demo.parquet
├── results/
│   ├── figs/                  # fig_*.pdf/png
│   ├── tables/                # tab_*.csv/tex
│   └── logs/                  # run.json, seeds, SPEC_ID, git-hash
├── paper/
│   ├── tex/                   # LaTeX source with fig/tab includes
│   └── appendix/              # A–F source
├── docker/
│   ├── Dockerfile
│   └── compose.yaml
```

```
├── ci/
│   ├── test_invariants.py     # mass non-creation; monotonicity
│   ├── test_placebos.py       # BH nulls; synthetic checks
│   └── test_reproduction.py   # hashes of figs/tables on synthetic
├── LICENSE
├── CITATION.cff
└── README.md
```

F.2 Environment & Setup

- **Python** ≥ 3.10 (lock via `pyproject.toml`; deps: numpy, pandas, scikit-learn, statsmodels, matplotlib, pyyaml).
- **Julia** ≥ 1.10 (deps: DataFrames, StatsModels, Optim).
- **Docker** (optional): containerized repro.

Install (Python):

```
pip install -e code/python
# or: pipx run pipenv install --dev
```

Install (Julia):

```
julia --project=code/julia -e 'using Pkg; Pkg.instantiate()'
```

F.3 One-Command Reproduction

```
# 0) Paths & seeds
cp config/paths.example.yaml config/paths.yaml   # point to your data/raw
export SD_SEED=20250901

# 1) Indicators
python code/python/scripts/build_indicators.py --config config/indicators.yaml

# 2) Threshold calibration (ROC/PR)
python code/python/scripts/calibrate.py --thresholds config/thresholds.yaml --horizon 8

# 3) Kink & IV for θ1 and Λ
python code/python/scripts/kink_iv.py   --spec config/instruments.yaml
python code/python/scripts/lambda_iv.py --spec config/instruments.yaml

# 4) Early warning & BH flags
python code/python/scripts/bh_detect.py --thresholds config/thresholds.yaml

# 5) MPC / Robust MPC
python code/python/scripts/run_mpc.py --policies config/policies.yaml --horizon 12

# 6) Figures & Tables
python code/python/scripts/make_figs.py   --out results/figs
python code/python/scripts/make_tables.py --out results/tables
```

Expected outputs (hash-stable on synthetic data):

- `results/figs/fig_phase_map.pdf` (SHA256 logged)
- `results/figs/fig_roc_pr.pdf`
- `results/tables/tab_kink_theta.csv` ($\hat{\theta}$, slope jump, CI)
- `results/tables/tab_lambda_paneliv.csv` ($\hat{\Lambda}$ profiles over horizons)
- `results/tables/tab_bh_performance.csv` (AUROC/AUPRC/Brier/ECE)
- `results/logs/run.json` (SPEC_ID, seed, git hash, environment)

F.4 Data Ingestion & Minimal Demo

- **Synthetic panel (included):** `demo_panel.parquet` and `text_sse_demo.parquet` match the schema in Appendix D and drive the full pipeline.
- **Real panel (user-supplied):** set paths in `config/paths.yaml`; ingestion scripts in `code/python/src/io_*.py` map source columns to Appendix D names/units.

Sanity checks:

```
pytest -q ci/test_invariants.py
pytest -q ci/test_placebos.py
```

F.5 Figure & Table Map (Paper ↔ Files)

Paper Ref	File	Generator	Notes
Fig. 1 Phase Map & BH Cube	`figs/fig_phase_map.pdf`	`make_figs.py --phase`	SAFE/CRITICAL/BH regions & trajectory
Fig. 2 ROC/PR	`figs/fig_roc_pr.pdf`	`make_figs.py --rocpr`	Star at best F_β
Fig. 3 Scenario Tree	`figs/fig_scen_tree.pdf`	`make_figs.py --scen`	Pruned branches & Pareto tags
Tab. 1 θ1 (Kink-IV)	`tables/tab_kink_theta.csv`	`kink_iv.py`	By type/sector
Tab. 2 Λ (Panel-IV)	`tables/tab_lambda_paneliv.csv`	`lambda_iv.py`	Dynamic profile $h = 0..8$
Tab. 3 BH Performance	`tables/tab_bh_performance.csv`	`bh_detect.py`	AUROC/AUPRC/Brier/ECE
Tab. 4 MPC Plans	`tables/tab_mpc_plans.csv`	`run_mpc.py`	Plan mix, budgets, violations

F.6 Continuous Integration (CI) Tests

- **Invariant checks:** in pure-conversion windows verify $\Delta \sum_\tau \sigma \le 0$.
- **Monotonicity:** spot-check $\partial \mathrm{SSI}/\partial \sigma \ge 0$, $\partial \mathrm{SSI}/\partial C \le 0$.
- **BH uplift:** $\Delta p = \Pr(Y \mid BH) - \Pr(Y \mid \neg BH) > 0$ with block-bootstrap CI.
- **Placebos:** random thresholds & unit placebos → AUROC ≈ 0.5.
- **Spec ledger:** every run must write `results/logs/run.json` with `SPEC_ID`; CI fails otherwise.

Run with:

```
pytest -q ci
```

F.7 Extending the Package

Add a new country/industry. Map source columns to the Appendix D schema, update `config/paths.yaml`, rerun F.3.

Add a new surplus type $\tau^\star$.

1. Define proxies for $\sigma_{\tau^\star}$ and $C_{\tau^\star}$ in `indicators.yaml`.
2. Register computation in `sd_indicators.py`.

3. Update `thresholds.yaml` so `max_τ` includes $\tau^\star$.

Add a new policy lever $u^\star$. Add elasticities $(\alpha_\tau, \beta, \gamma)$ in `policies.yaml`; `sd_mpc.py` will ingest automatically.

F.8 Provenance, Licensing, Citation

- **Code:** MIT. **Text:** CC BY 4.0. **Data:** according to original sources.
- **Model card:** each run logs `SPEC_ID`, data window, SSE pipeline version, ROC/PR scores, and any overrides in `results/logs/run.json`.
- **Citation (`CITATION.cff`):** authors, title, year, DOI/URL (to be assigned).

F.9 Open Problems (Research Roadmap)

F.9.1 Theory & Geometry

1. **Global bifurcation with switching (threshold hysteresis).**
   Goal: characterize basin volume as a function of (k, b_1, Λ).
   Start: piecewise-smooth dynamics + Conley index.

2. **Networked conversion.**
   Goal: identify a minimum-cut set of conversion links that prevents crisis routing.
   Start: cut/flow methods with submodular relaxations.

F.9.2 Measurement & SSE

3) **Manipulation-resistant SSE.**
Goal: keep metric–event divergence ≤ 5pp under source reweighting and messaging campaigns.
Start: multi-view entropy + adversarial training.
4) **Cross-lingual SSE.**
Goal: correlation ≥ 0.8 across major languages.
Start: multilingual embeddings with topic alignment.

F.9.3 Identification

5) **Dynamic Λ under latent confounding.**
Goal: bias bounds in panels with event studies.
Start: proximal IV / front-door designs.
6) **θ1 under sample selection near the threshold.**
Goal: selection-corrected RKD/kink estimators.
Start: Heckman-style RKD with selection equation.

F.9.4 Control & Governance

7) **Distributionally robust MPC (Wasserstein + chance constraints).**
Goal: ≥30% reduction in out-of-sample BH violations vs. nominal MPC.
Start: scenario bundles + dual reformulations.
8) **Equity-aware controllers.**
Goal: non-emptiness conditions for feasible sets with Roemer-style exploitation bounds.
Start: bilevel feasibility with convex surrogates.
9) **Human-in-the-loop dashboards.**
Goal: pass Goodhart-resilience A/B tests.
Start: bandit-style policy messaging.

F.9.5 Externalities & Sustainability

10) **Ecological surplus E.**
Goal: extend the phase map with resource/carbon constraints while retaining usability.
Start: stock–flow ledgers mapped into CRP/SSE.

F.9.6 Computation & Tooling

11) **Auto-spec search.**
Goal: match hand-tuned AUROC within 0.02 and ECE ≤ 5pp across domains.
Start: automated SPEC_ID exploration regularized by IRM/robustness scores.
12) **Streaming pipeline.**
Goal: recover out-of-sample calibration within four periods after a structural break.
Start: CUSUM + online grid search with audit logs.

F.10 Reproduction Checklist (for Reviewers)

- Record `SPEC_ID`, random seed, and environment hash.
- Run `make all` or the stepwise commands in §F.3 to generate figures/tables.
- Pass CI tests (invariants, placebos, BH uplift).
- Report robustness using at least **two** indicator pipelines **and two** threshold bundles.
- Attach `results/logs/run.json` and SHA256 checksums for main figures/tables.

End of Appendix F.

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Disclaimer

This book is the product of a collaboration between the author and OpenAI's GPT-5 language model. While every effort has been made to ensure accuracy, clarity, and insight, the content is generated with the assistance of artificial intelligence and may contain factual, interpretive, or mathematical errors. Readers are encouraged to approach the ideas with critical thinking and to consult primary scientific literature where appropriate.

This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.

I am merely a midwife of knowledge.